SIMPLIFIED PARTICULATE TRANSPORT MODEL (SIMPTM)

Version 3.2
December 1998

by
Roger C. Sutherland, PE
and
Seth L. Jelen, PE

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We Think the World of Water
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FOREWORD

Conceptual development of the SIMPTM package began in 1973 at the University of Maryland during my graduate studies in water resources engineering. My master's dissertation entitled, "A Mathematical Model for Estimating Pollution Loadings and Removals for Urban Streets" (Sutherland, 1975) laid the groundwork for the Management of Nonpoint Pollution (MUNP) Model (Sutherland, Brazil and Mades, 1977) which was written during the evening hours while working at Harza Engineering in Chicago, Illinois. At that time, Mr. Dean Mades wrote the first version of the RAINEV program and Mr. Larry Brazil worked on the first storage/treatment block.

An expanded version of the MUNP model was created and applied throughout Illinois in 1978 as part of a statewide 208 study that was obtained immediately after my joining CH2M HILL's Redding, California office. The results of this project laid the foundation for the USEPA-funded National Urban Runoff Project in Champaign, Illinois conducted by the Illinois State Water Survey (Terstriep, Bender and Noel, 1982).

The initial washoff subroutine for the Particulate Transport Model (PTM) was written (Sutherland and Ellis, 1978) in an attempt to gain funding from the USEPA to develop an alternative approach to EPA's Storm Water Management Model (SWMM). SWMM was quite popular at the time but it was not very effective in predicting urban storm water pollutant discharges (Ellis and Sutherland, 1979). Although unsuccessful with EPA, some funding was obtained from the USGS's Reston, Virginia office and PTM was created (Sutherland, Alley and Ellis, 1980).

Version 1.0 of the Simplified Particulate Transport Model (SIMPTM) was written in 1980 and applied to over 57,000 acres of urban and urbanizing lands as part of a comprehensive urban storm water management program effort for the Washoe County Council of Governments in Reno, Nevada (Sutherland, 1982). Dr. Robert Pitt worked as a subconsultant on the project and provided considerable input to the overall structure and several routines in the program such as the street cleaning component.

The conversion of the SIMPTM package to the PC was done by Mr. Dennis Green in 1987. Since that time, Mr. Green has rewritten almost all of the original code in a major effort to bring the SIMPTM package up to Fortran 77 standards.

The next version of the program (*i.e.* Version 2.0) was created by Mr. Green in 1989. The underlying motivation for the tremendous expansion of SIMPTM's capabilities was both the City of Portland, Oregon's Columbia Slough Water Quality Management project in which OTAK, Inc. acted as a subconsultant to Brown and Caldwell and the recent research by Dr. Robert Pitt in the area of small storm hydrology. In a last minute March 1989 effort, Mr. Green recoded the old storage and treatment block of SIMPTM so that it would function as an on-site retention component.

In 1990, Mr. Seth Jelen joined OTAK, Inc. and began working on various water quality projects including the first phase of Portland, Oregon's Combined Sewer Overflow (CSO) Abatement Project. His skills in programming led to several changes in the code that made it more efficient in its execution and the release of Version 2.1 in November 1990.

In March 1992, Version 3.0 was released. Due to significant advances in spreadsheet technology and our continued work on the Portland CSO project which required a direct linkage with the city's version of USEPA SWMM, the

output export option to spreadsheets using quote delimited files was created. The other output processors PTMOUT and SUMMARY were retained as optional choices.

In October 1993, Mr. Gregory Kurahashi, the "K" in OTAK, left the company and started his own company, Kurahashi & Associates, Inc. (KAI). Greg asked me and the entire water resources group that I had built with his help to join him. Several chose to join KAI including myself, Mr. Seth Jelen and Mr. Habib Matin. Others chose to stay including my good friend and colleague Dennis Green. KAI purchased the rights to SIMPTM which led to the release of Version 3.1 in December 1993.

In October 1998, Mr. Jelen, Mr. Matin and I created a new company called Pacific Water Resources, Inc. (PWR). We sold our ownership interest in KAI and negotiated an agreement that included the rights to SIMPTM. The current version (*i.e.* Version 3.2) of the model was released in December 1998. This version abandoned the optional output processors PTMOUT and SUMMARY. But most importantly, it includes a unique equilibrium accumulation function and a wet weather accumulation function that has been referred to as wet weather washon. These behaviors were discovered by Mr. Jelen as part of our continued work on subsequent phases of the Portland CSO program. I have jokingly referred to these discoveries as "Sutherland's Flux Capacitor" named after the hit movie Back to the Future. In the movie, the "flux capacitor" is the final ingredient that allows space time travel. So you see, these discoveries have simply given SIMPTM the final ingredient needed to continuously simulate in an accurate manner the event-to-event washoff of storm water pollutants season after season, year after year.

On a more personal note, I would like to thank everyone who has helped me with this program package throughout the years. The hours that each of you spent without compensation, but because of our friendship and mutual interest in this topic are gratefully appreciated. I would like to specifically thank Mr. Gregory Kurahashi and Mr. Nawzad Othman for their encouragement and financial support of this endeavor and my partners Mr. Habib Matin and Mr. Seth Jelen for their continued support of this work in progress.

And finally, I would like to rededicate this current version of the modeling package to Dr. Richard H. McCuen, my thesis advisor at the University of Maryland. Dr. McCuen instilled within me (his first graduate student) the belief that properly formulated, verified and calibrated mathematical models can reproduce within reasonable error the complicated responses of nature. But even more importantly, these models can tremendously increase our knowledge of the systems we attempt to emulate. The SIMPTM package is a demonstration of these beliefs.

My hope is that the pendulum between model creation and data collection will now swing back to the balance point and that funding will become available to conduct a detailed model calibration and regional parameter extraction on the Nationwide Urban Runoff Program data collected in the early 1980's and the newer NPDES data collected in the early 1990's. Let us explore the possibilities of high-efficiency street sweeping and other promising technologies. Let us determine why some control measures may work well in one part of the country and not very well in other parts. Let us determine what we can do to solve this problem. The tools are now available. We owe this to a nation of people who once again through their Congress has reaffirmed their beliefs that we can and we must control the quality of our urban runoff.

Roger C. Sutherland, M.S.C.E., P.E.
Vice President
Director of Water Resources Engineering
March 1999

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SECTION 1

INTRODUCTION

1

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SECTION 1
INTRODUCTION

1.1 Installation

The Simplified Particulate Transport Model (SIMPTM) Package is a group of PC-based programs that simulate pollutant loadings transported by urban storm water on an event-by-event basis. SIMPTM requires an IBM AT or fully compatible with a math co-processor. The programs were compiled with the Lahey Fortran 77 Compiler, Version 5.00. If an error occurs, the program looks up the message in the error file `f77l.eer`. Without it, just a number is displayed. Appendix B - Runtime Error Messages, interprets these numbers should the file be unavailable. To change the language used, copy the desired file `language.eer` to `f77l.eer`. French, Spanish, Italian, German, Dutch, and the default, English, are provided.

The master diskette contains the entire package in compressed format. To install the SIMPTM package on your X: drive, go to the source diskette directory and type (`install x:`). It's that easy. Installation will proceed in three steps:

1. Executable files will go to `x:\simptm\` directory.

2. Sample files will go to `x:\simptm\samples\` directory. Though they are not needed for program execution, they are well worth having as input templates. Also, they are needed for the test run.

3. A test run ensures that installation proceeded correctly, and will introduce the package. The programs will process the sample files, creating output that can be compared with the given files.

Should any optional files be wanted later, re-INSTALL the package or manually re-explode the desired files.

1.2 SIMPTM Package Overview

The Simplified Particulate Transport Model package consists of the stand-alone programs RAINEV and SIMPTM. RAINEV, the rainfall evaluator, performs the first step in simulating storm water pollution — rainfall characterization. It groups a continuous hourly precipitation record into significant events and tabulates the starting date and time, duration, depth, time since the last event ended, average intensity and maximum hourly intensity for each one. Monthly and annual averages of these can be grouped into tables either by month, parameter, or year. In addition, RAINEV fits each event rainfall distribution to a rectangle, triangle, and trapezoid — used for SIMPTM. Information for each event is recorded in a log file which can be used by SIMPTM, re-read and re-summarized by RAINEV or easily imported into spreadsheets or databases for further analysis.

RAINEV accepts hourly precipitation data in two formats: that used by the National Weather Service and having 24 hours of data on 2 lines, and a second 1-hour-per-line format used by some automatic recording gages. Output is even more flexible. Summaries can be spreadsheet-ready "numbers" or "quote-delimited" files or simple easily read ASCII files. The log file, listing information for every significant rainfall event, can be re-read and re-summarized in a different format, saving the time used to identify events from raw data.

RAINEV provides valuable information about rainfall events to support not only SIMPTM runs but also NPDES stormwater permitting. Rainfall event characteristics can be tabulated for pollution reduction facility sizing, while the summary tables can assist in arranging a "representative year" which reflects the

distribution of rainfall observed in long records. Representative washoff simulations can thus be done using one year of data, not the entire record.

SIMPTM simulates event-by-event stormwater discharges and pollutant loadings from developed lands using published, physically-based sediment transport equations. Results are all spreadsheet-ready as "numbers" or "quote-delimited" files for easy post-processing analysis of data. The overall effectiveness of strategies such as street sweeping, sediment trapping catchbasins, and catchbasin cleaning are output by SIMPTM simulations, rather than user-supplied inputs as with most stormwater quality models. SIMPTM uses Pitt's small-storm hydrology work to model initial rainfall losses far better than SCS curve numbers would, and accounts for sediment deposition and re-suspension processes.

For sediment transport, SIMPTM uses equations published by Yalin, Einstein and Foster and Meyer and an average trapezoidal hydrograph at a stormwater inlet to simulate the washoff of accumulated sediment from paved areas. This has been shown by Ellis and Sutherland (1979) to be much more accurate than the empirically-based exponential washoff used in EPA's SWMM model. SIMPTM compares the fraction of supply available for transport with the shallow-flow transport capacity from Shield's Diagram and considers the armoring of smaller particles by larger overlying ones.

Many basin-wide planning projects will require simulating unit or per-acre runoffs and pollutant washoffs for each of many contributing land uses which must then be combined. Using the spreadsheet-ready output files from SIMPTM this can be easily accomplished in any spreadsheet using matrix multiplication to weigh output results by contributing areas for each of any number of subbasin areas. Thus subbasin results can be grouped into basin-wide predictions.

Many input parameters can be measured or easily estimated from field-survey results, USGS topographic maps, aerial photos, or drainage system inventories. However, others are not well known or have not been measured to a great extent. The Nationwide Urban Runoff Program (NURP) has extensive data from some thirty sites across the country. Two sites in Bellevue, Washington have provided valuable information to calibrate the model for the Pacific Northwest, but other regions might be better served by other sites. The following is suggested in calibrating the model:

1. Match runoff volume by adjusting the effective impervious area and the impervious area variable loss parameters. Establish the relation between rainfall and runoff durations.

2. Match total solids (TS) or total sediments accumulation and smaller-storm washoffs by adjusting the impervious accumulation and availability. Pervious area contributions would be significant only during larger events. Sufficient data of large-storm washoffs might support inclusion of pervious runoff which could carry extra pervious-area sediment observed to washoff.

3. Match other pollutants in either SIMPTM by adjusting their mass-to-mass pollutant strengths for each particle size group. Compare TS-pollutant ratios on paved surfaces with those washing off and in sediment traps.

4. Verify parameters using data from a different period of record.

5. Change parameters to reflect observed differences in land use characteristics between the calibrated and to-be-simulated basins. In the end, good judgement must assess the results.

1.3 Program Files

The programs look for input files and write output files with certain default extensions. Figure 1.1 illustrates how the program and their files relate to a project:

RAINEV

*.IN	Instruction File
*.PRN	Event Summary Output
*.RN	Event Log for input by RAINEV or SIMPTM
*.NWS	Hourly Rainfall Data when NWS input format chosen
*.DAT	Hourly Rainfall Data when 1-hour-per-line format chosen

SIMPTM

*.LU	Land Use File
*.PRN	Output file for all options: short-form and both long-forms *.RN Event Log and *.POL Pollutant and Option Files are named in LU file

1.4 Program Execution

Sometimes it is nice to be prompted for information and supply needed filenames interactively. In this case, all filenames are separately asked for. At others, it is inconvenient to repetitively type the same names during a calibration. Fortunately, the programs may be invoked in three ways:

Interactively:

SIMPTM [ret]

Enter the program name at the prompt. It will then ask for the names of the required files, including, for SIMPTM, the RN and, if used, POL filenames.

Batch mode:

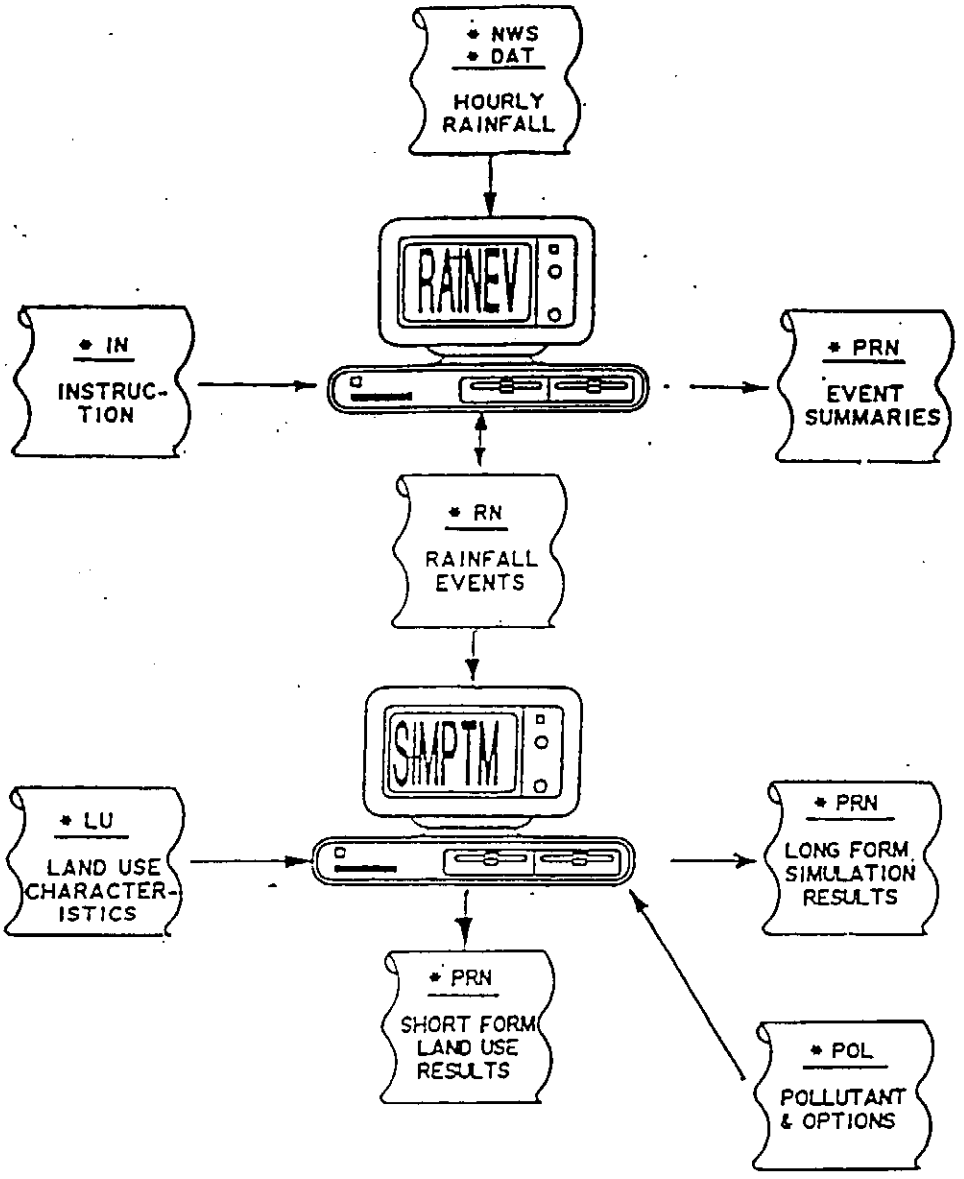
SIMPTM < filename[.ext] [ret]

Enter the program name followed a "<" and then by the name of an ASCII file listing the filenames asked for by the program in interactive mode. For SIMPTM, this includes the RN and, if used, POL filenames. Refer to the DOS documentation for further information about redirecting files.

Command Line Mode:

SIMPTM filename[.ext] [ret]

Enter the program followed by the filename of the land use file. The default extension is used unless another is given. The remaining files will share the same filename but use their own default extensions. For SIMPTM, the RN and, if used, POL filenames are read from the Land Use File. If the input file is not found the program requests a new one; if the output file already exists, confirmation to replace it is requested.


* NWS
* DAT
HOURLY RAINFALL
RAINEV
* IN
INSTRUC-TION
* PRN
EVENT SUMMARIES
* RN
RAINFALL EVENTS
SIMPTM
* LU
LAND USE CHARACTER-ISTICS
* PRN
LONG FORM SIMULATION RESULTS
* PRN
SHORT FORM LAND USE RESULTS
* POL
POLLUTANT & OPTIONS

SECTION 2

RAINEV

2

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SECTION 2
RAINEV

RAINEV is a rainfall analyzer which performs the first step in simulating storm water pollution — rainfall characterization. It divides an hourly precipitation record into significant rainfall events and calculates the starting date and time, duration, depth, time since the last event ended, average intensity and maximum hourly intensity for each one. Monthly and annual averages of these can be grouped into tables by month, parameter, or year. In addition, RAINEV fits each event rainfall distribution to a rectangle, triangle, and trapezoid — used for SIMPTM. Information for each event is recorded in a log file which can be used by SIMPTM, re-read and re-summarized by RAINEV or easily imported into spreadsheets or databases for further analysis.

RAINEV accepts hourly precipitation data in two formats: the National Weather Service 24-hour, 2-line record and a 1-hour-per-line format used by some recording gages. This format is documented in Section 4. Output is even more flexible. Summaries can be spreadsheet-ready "numbers" or "quote-delimited" files or simple easily read ASCII files. The log file, listing information for every significant rainfall event, can be re-read and re-summarized in a different format without having to re-identify events from raw data.

RAINEV provides valuable information about rainfall events to support not only SIMPTM runs but also NPDES stormwater permitting. Also, rainfall event characteristics can be tabulated for pollution reduction facility sizing, while the summary tables can assist in arranging a "representative year" which reflects the distribution of rainfall observed in long records. Representative washoff simulations can thus be done using one year of data, not the entire record.

2.1 Event Location

In evaluating rainfall, the object is to eliminate what is insufficient to produce observable runoff. To do this, RAINEV separates rainfall into a series of events. Figure 2.1 illustrates how this is done:

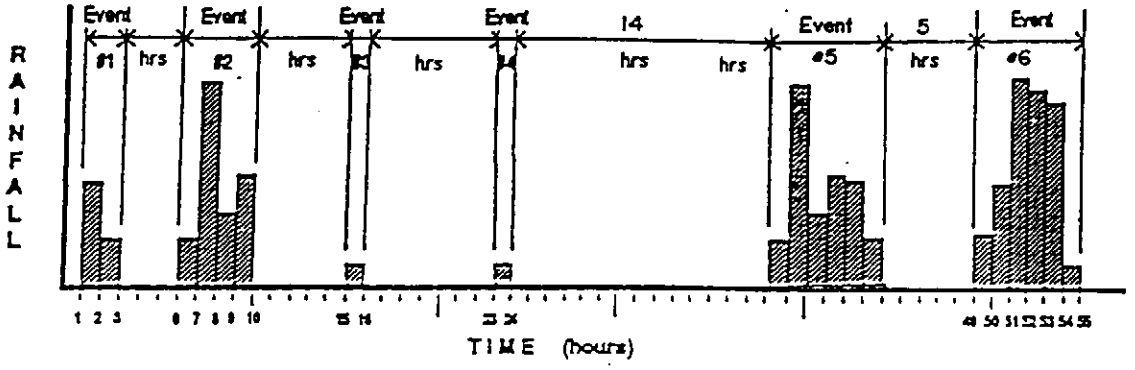
1. Hourly rainfall below a TRACE is ignored. This amount of rainfall could fall continuously yet entirely infiltrate or evaporate; even if surfaces were already wet, it would still be lost. Thus only hourly rainfall greater than this is worth considering.

2. Events are separated by a LAPSE time sufficient to dry out the impervious surfaces. Obviously, this varies with location, season, weather and surface texture, but RAINEV accepts only one value per simulation. Although larger times might produce more statistically independent events, the events would include large dry periods in their rainfall distributions and be very ill-suited for SIMPTM modelling purposes.

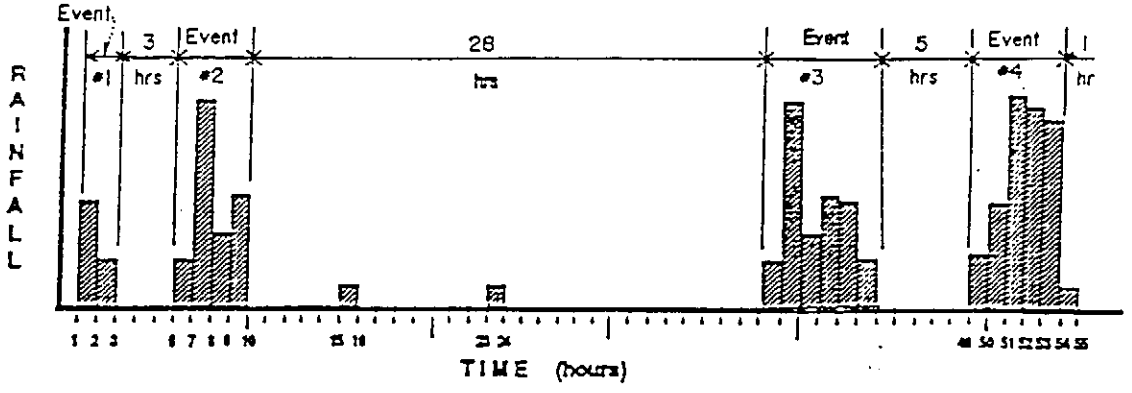
3. Rainfall THRESHolds for 1, 3, and 6 hours eliminate events which would produce no surface runoff. Sizeable initial losses from infiltration, evaporation, and particularly surface depression storage exist even on impervious surfaces, although far less than on pervious land. These must first be satisfied before any excess precipitation is available for runoff. Common sense would indicate that the threshold of rainfall needed for any significant runoff should increase with time, but at a decreasing rate. While initial losses are being satisfied, initial infiltration is depleting surface storage, leaving still more to fill. Impervious surface loss goes mostly to fill surface depressions and provide for initial infiltration before surface runoff begins. A generally accepted value for has been 0.1 inch of water over the impervious area, based on traditional flood hydrographs and peak flow analyses. However, research conducted on small storm hydrology shows that this is too high for small storms. Arnell (1982) observed initial losses for mixed urban watersheds in Germany to vary from 0.015 to 0.03 inch for watersheds having impervious fractions ranging from 19 to 46 percent. Pratt and Henderson (1981) examined several European impervious test plot data sets and found values to generally decrease from 0.06 to 0.02 as land slope increased from 0.5 to 10 percent.

FIGURE 2.1
Rainfall Event Location

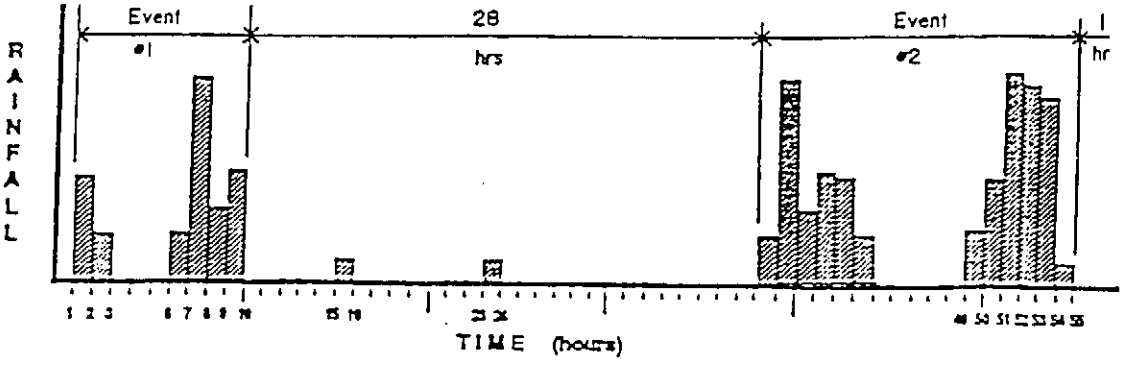
Rainfall Events: Lapse = 1 hour, Trace = 0



Rainfall Events: Lapse = 3 hours, Trace > 0



Rainfall Events: Lapse = 6 hours, Trace > 0



In experimental rainfall tests on street surfaces in Toronto, Canada, Pitt (1987) observed initial losses between 0.012 and 0.043 inches to increase with pavement roughness. Also, significant variable losses suggested that about 0.05 inches of rainfall would result in only 0.01 inches of runoff. Rainfall intensity did not seem to affect these initial impervious-area losses. However, all tests lasted less than 2.12 hours and involved intensities greater than 0.11 inches per hour.

The following are reasonable first estimates for SIMPTM simulations:

Parameter	Value
LAPSE	4 hours
TRACE	0.01 inches
THRESH1	0.02 inches
THRESH3	0.04 inches
THRESH6	0.07 inches

Experiment and see how these impact event statistics.

2.2 Rainfall Distributions

RAINEV fits the rainfall distribution from each event to three different forms. The two-parameter rectangular is perhaps the simplest to visualize, but the three-parameter triangular and four-parameter trapezoidal contain increasing information on the event and reflect the rainfall distribution increasingly well. Each form is illustrated in Figure 2.2. Although SIMPTM uses the trapezoidal pattern exclusively, other uses may require one of the others, so all are provided. In fitting the rainfall, characteristics of the distribution are equated with those of the desired pattern. All fits will be worse as events lengthen and increasing dry periods are averaged in, so don't let lapse get too long.

The rectangular distribution only needs two parameters to reflect its base and height:

- The duration determines the length of the base.
- The depth determines the "area" of the shape, and thus the height.

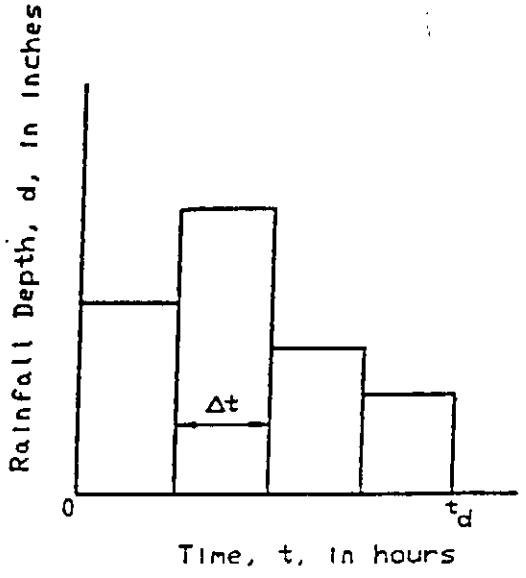
The triangular distribution requires an additional parameter to reflect the time-to-peak:

- The centroid of the rainfall determines the time-to-peak. This is the average of the time from event start weighted by the proportion of the total precipitation that fell then.

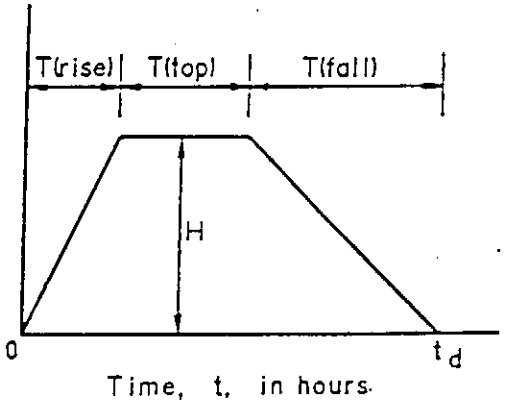
The trapezoidal distribution requires still one more parameter:

- The second moment of the rainfall reflects how concentrated or dispersed the rainfall is in time and provides information for the trapezoid top width.

This trapezoidal distribution requires more computation, as two non-linear equations for the centroid of rainfall and the second moment must be solved simultaneously to arrive at the rise-, top-, and fall-times. RAINEV uses the Newton-Raphsen Method to numerically solve this system.



EXAMPLE HYETOGRAPH



TRAPEZOIDAL REPRESENTATION OF A RAINFALL HYETOGRAPH

FIGURE 2.2

Trapezoidal Representation of a Rainfall Event

2.3 Execution Options

RAINEV provides four outputs: the log file (*.RN), which lists all significant rainfall events, and three different monthly and annual summaries arranged by month, parameter, and year. Event information includes the starting date and time, duration, depth, time since the last event ended, average intensity and maximum hourly intensity. Also, the log file reports information fitting each rainfall distribution to rectangular, triangular, and trapezoidal patterns. Note that it can be read and re-summarized by RAINEV, saving considerable time over re-processing the hourly rainfall data.

The log file reports the following:

Heading	Description
Evt	Events since the first of the current simulation year
CalHr	Hours since the first of the current simulation year (0 to 8759 or 8783)
Mo	Month
Da	Day
Yr	Year
Hr	Hour (0 to 23)
Dur	Hours of Rainfall
Depth	Inches of Rainfall
TDry	Hours since the end of the previous rainfall event (Lapse)
Int	Average inches per hour of rainfall
PMax	Maximum inches per hourly rainfall

These additional parameters are used to fit the rainfall distribution to triangular and trapezoidal patterns:

Heading	Description
TRise	Hours duration of trapezoid rise
TRun	Duration of trapezoid peak
TFall	Duration of trapezoid fall
H	Height of the trapezoid (inches per hour)
T	Hours-to-peak of triangular distribution

Annual totals and record averages are provided for six monthly statistics: number of events, total duration and depth, and average TDry, PMax, and Int. Although incomplete record years can be used, the entire-record averages will not be accurate for every month since the number of whole and partial years is used. These summaries are particularly useful when reporting general rainfall characteristics, combining different months to form a representative record, and checking the synthesis results. The information contained in each is identical, but graphing results are very different-looking:

The summary by month allows every parameter's departure from the record average to be graphed at once. Those years whose month is most representative of the months on record are clearly shown as dips in the lines. Figure 2.3 illustrates this process. When repeated 11 more times, this process identifies the months whose event records can be combined to form the "synthetic year" record.

The summary by parameter allows easy tracking of the annual variation of parameters for the record years. Several years seasonal variations are easily compared at once, though only parameter-by-parameter. Figure 2.4 illustrates how this arrangement can show how well synthetic year represents the record averages.

FIGURE 2.3

Selecting the Best Historical July for Use in the Representative Dry Season Record

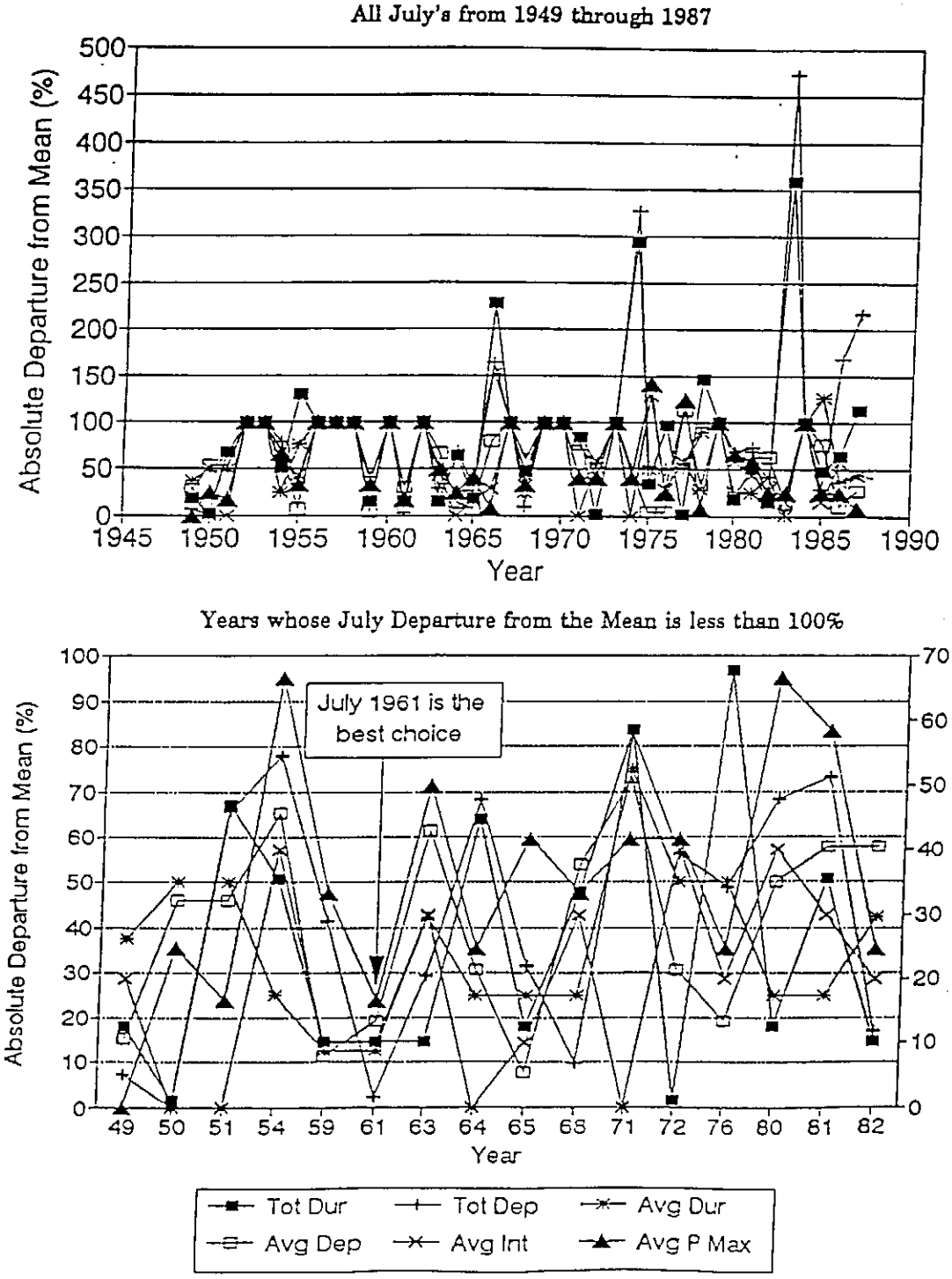
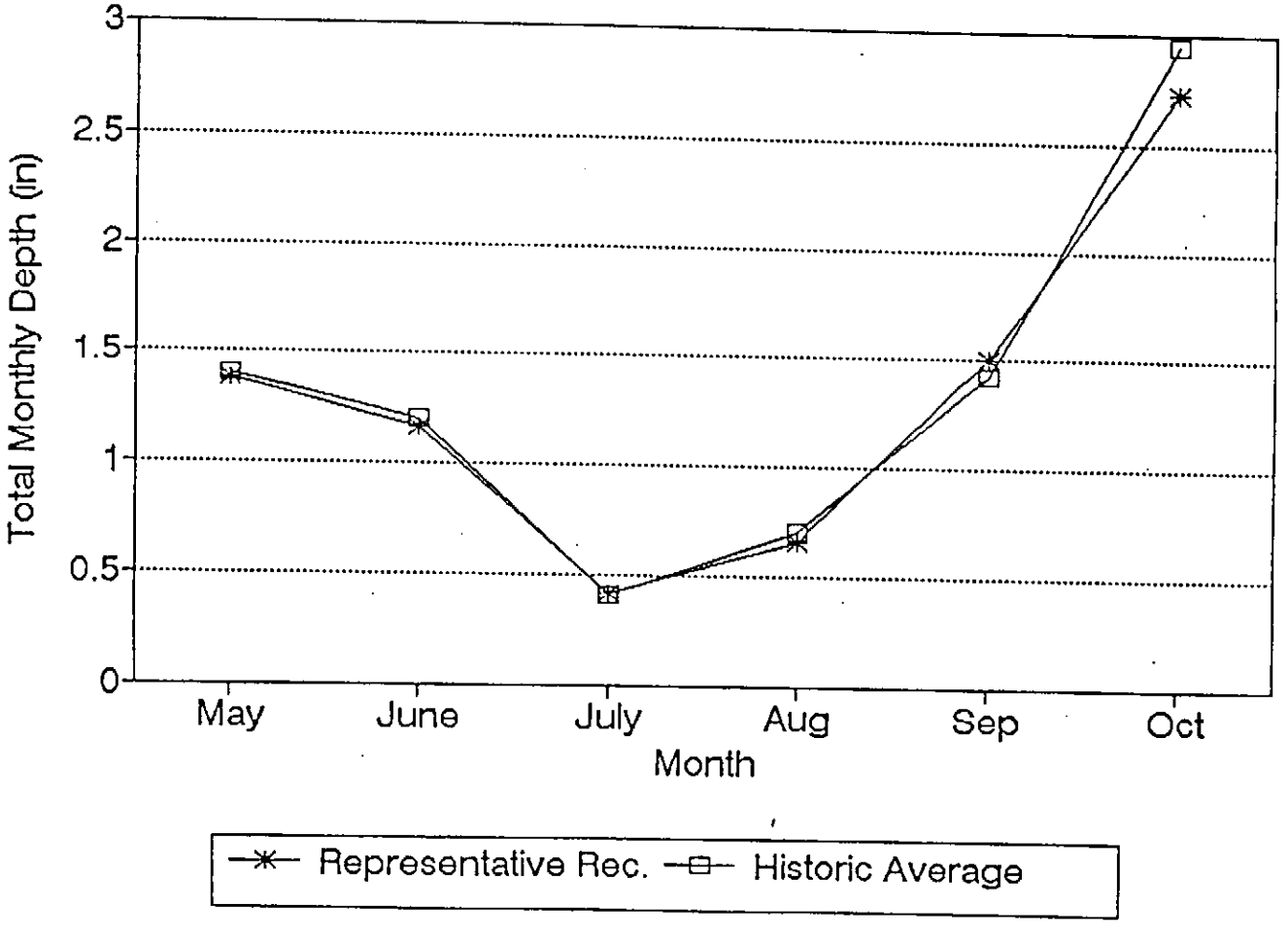


FIGURE 2.4

Representative Dry Season Rainfall Record Check



The summary by year allows an entire representative year to be selected when it is desired to use an actual entire-year record instead of aggregate monthly records. This arrangement also tracks parameters' variation through the months, and by looking at many years one-by-one the best one can be selected. Although less representative than the synthetic year, a representative year might be required for some functions.

After the event selection criteria have been selected, RAINEV need only be run once. Afterwards, the rainfall event log file can be trimmed as needed or arranged into a synthetic record. The log file can then be read and re-summarized by RAINEV to check the representative year.

SECTION 3

SIMPTM

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SECTION 3
SIMPTM

SIMPTM simulates event-by-event stormwater discharges and pollutant loadings from developed lands using published, physically-based sediment transport equations. Results are all spreadsheet-ready "numbers" or "quote-delimited" files for easy post-processing analysis of data. The overall effectiveness of strategies such as street sweeping, sediment trapping catchbasins, and catchbasin cleaning are output by SIMPTM simulations, rather than user-supplied inputs as with most stormwater quality models. SIMPTM uses small-storm hydrology to model initial rainfall losses far better than would SCS curve numbers and accounts for sediment deposition and resuspension.

For sediment transport, SIMPTM uses equations published by Yalin, Einstein, Foster and Meyer and an average trapezoidal hydrograph at a stormwater inlet to simulate the washoff of accumulated sediment from paved areas. This has been shown by Ellis and Sutherland (1979) to be much more accurate than the empirically-based exponential washoff used in EPA's SWMM model. SIMPTM compares the fraction of supply available for transport with the shallow-flow transport capacity from Shield's Diagram and considers the armoring of smaller particles by larger ones shadows.

Many input parameters can be measured or easily estimated from USGS topographic maps, aerial photos, field-survey results, or drainage system inventories. However, others are not well known or have not been measured to a great extent. The Nationwide Urban Runoff Program (NURP) has extensive data from some thirty sites across the country. Two sites in Bellevue, Washington have provided valuable information to calibrate the model for the Pacific Northwest; other regions may be better served by other sites. to calibrate SIMPTM:

1. Match runoff volume by adjusting the effective impervious area and the impervious area variable loss parameters. Correlate duration of runoff to rainfall.

2. Match total solids (TS) or total sediments accumulation and smaller-storm washoffs by adjusting the impervious accumulation and availability. Pervious area contributions would be significant only during larger events. If large-storm washoff data supports, include pervious runoff to carry extra pervious-area sediment observed.

3. Match other pollutants with SIMPTM by adjusting their mass-mass pollutant strengths for each particle size group. Compare pollutant-TS ratios on paved surfaces with those of washoff and trapped sediment.

4. Verify parameters using data from a different period of record.

5. Change parameters to reflect observed differences in land use characteristics between the calibrated and to-be-simulated basins. In the end, use good judgement to assess the results.

3.1 Simulation Options

SIMPTM reads rainfall event data only when prepared as a Log File by RAINEV, but three different output formats are available: two long-form reports detail nearly all simulation variables or just a subset of accumulation over time, and one short-form report lists event resulsts with any pollutants arranged in several ways.

The long-form output provides spreadsheet-ready information so that accumulation of sediment on paved surfaces or sediment traps can be compared with measured data, and catchbasin cleaning and street sweeping results tracked. The information is in five sets of intermixed records. Note that the long-forms report only unit or per-acre TS results. The following information is common to all sets:

Num	Number of the washoff, street sweeping or trap cleaning event
Key	Key to data record described below (0-4).
CalHr	Calendar hour from the start of the year (0 to 8759 or 8783)
Yr	Year of record
Date	Month, day and hour (0 to 23) of record

Though intermixed, the five data sets are easily separated in a spreadsheet by sorting first by key, then by number. The following records are listed by key:

0. Remaining columns hold event washoffs:

Wash	Total pounds per acre of sediment washoff for all eight size groups
1—8	Pounds per acre of sediment washoff for each of eight size groups
Run	Acre-feet of total runoff for the event
SFEvt	Fraction of impervious accumulated sediment available for washoff

1. Remaining columns hold impervious accumulations:

TotSol	Total accumulation for all eight size groups. Units are pounds per curb mile or paved acre, depending on whether curb-and-gutter or overland flow, respectively, was simulated.
1—8	Accumulated sediment in each of eight size groups
CBtot	Total accumulation for all eight groups in sediment traps (lbs/acre)

2. Remaining columns hold event street sweepings:

Swept	Total pounds per acre swept for all eight size groups
1—8	Pounds per acre swept for each of eight size groups

3. Remaining columns hold catchbasin trap accumulations:

CBtot	Total pounds per acre accumulated for all eight size groups (same value appears above with impervious accumulation)
1—8	Pounds per acre of each size group stored

4. Remaining columns hold cumulative catchbasin cleanings:

TotcIn	Total pounds per acre cleaned from catchbasins from ALL cleaning events for all eight size groups
1—8	Pounds per acre cleaned for each of eight size groups for ALL cleaning events

Although this long-form output has its uses, the other two forms are more commonly used. The accumulation-only long-form output is available to compare modeled to observed accumulation.

For most other uses, the short-form output presents needed information in the less space. Also, this form can report runoff only or any pollutants in addition to total solids, and can include event mean concentrations with the total loads. Multiple-land-use files are combined external to the SIMTPM package using spreadsheet-based calculations to formulate composite basin results.

The number of columns depends upon what is simulated, but the following columns are common to all short-form output files:

Evt	Number of events since the simulation began
CalHr	Hours since the start of the current simulation year
Yr	Current simulation year
Date	Month, date and time (0 to 23) of this event
Dur	Hours of RAINFALL
Depth	Inches of RAINFALL
TDry.	Hours since the end of the last RAINFALL event
PMax	Maximum inches of hourly RAINFALL
RunD	Hours of RUNOFF
Acft	Acre-feet of RUNOFF volume
CFS	average event RUNOFF flow

When sediment washoff is simulated, additional columns list sediment washoff in pounds and pollutant washoff, if simulated, in the units given (usually pounds). When EMC's are requested, they will be in mg/L (and pollutants MUST be in pounds). Finally, pollutant washoff calculations can be deferred to later, external calculations by simulating only sediment washoff for each of eight size groups. Experiment with the various options and see how useful each can be.

3.2 Runoff Hydrograph

SIMPTM transforms the trapezoidal distribution obtained from RAINEV into a trapezoidal hydrograph by correlating duration of rainfall and runoff and transforming the rainfall depth into runoff volume using equations developed by Pitt (1987). After subtracting initial and variable losses, the excess precipitation is weighed by the effective impervious area (EIA) to give the runoff volume.

SIMPTM correlates runoff to rainfall durations:

$$T_{run} = A_{run} + B_{run} * T_{rain} \qquad \textit{[eq 3-1a]}$$

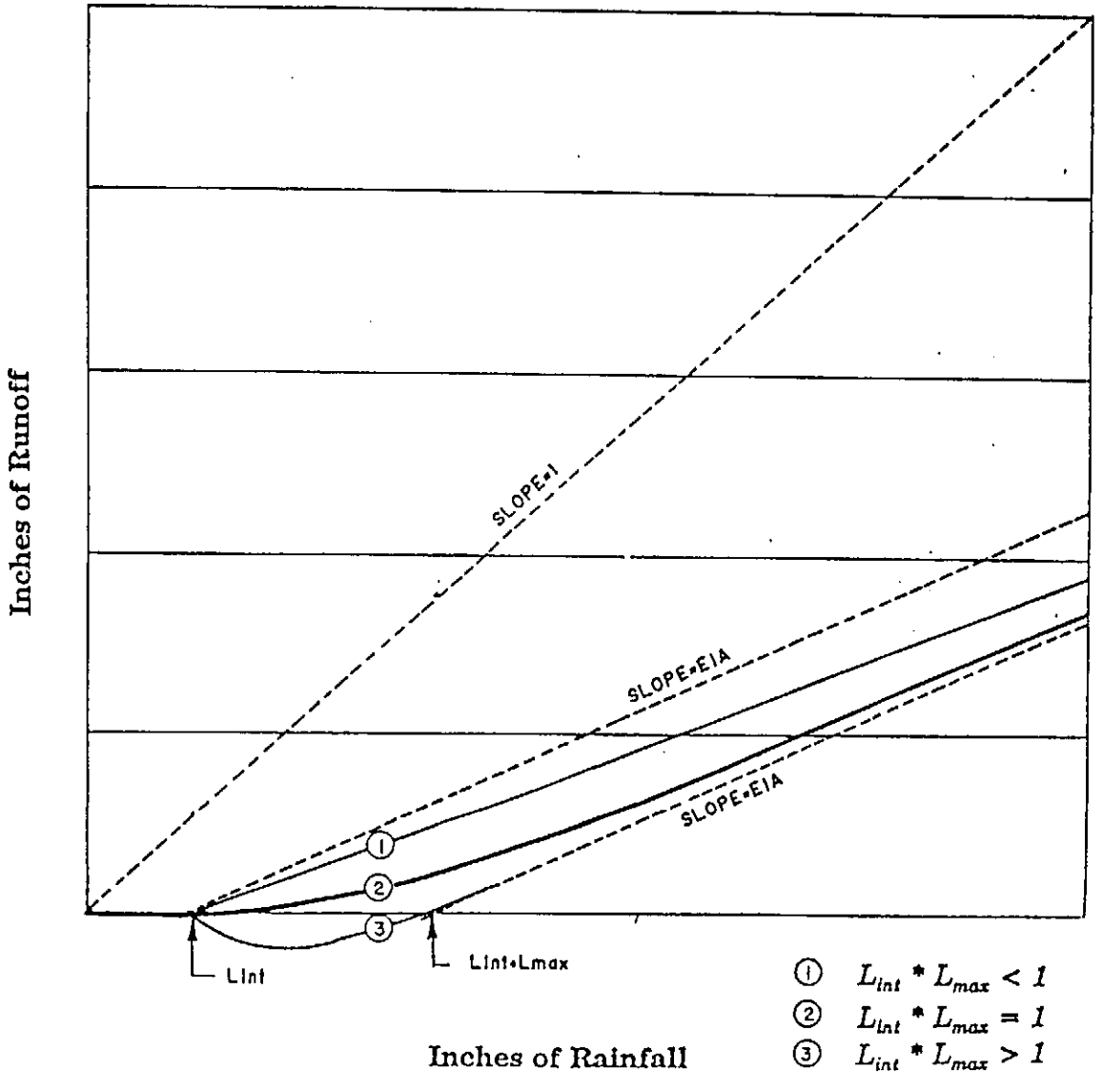
The two runoff coefficients Arun and Brun will vary by climate and basin geometry and should be based on regression of observed data. The default values (-0.6 and 1.1, respectively) were obtained from an set of over 400 observations on two NURP basins in Bellevue, Washington (Pitt, 1985). However, considerable variability exists between regions: 0.9 and 0.98, respectively, were obtained from an extensive Toronto, Canada database (Pitt and McLean, 1986; Pitt, 1987).

SIMPTM can model runoff from both impervious and pervious areas, the latter primarily to provide for a pervious washoff contribution. Losses from both areas' rainfall Depth have two parts: a constant initial loss (Lint) and a variable loss (Lvar). The excess precipitation weighted by EIA yields Runoff volume in inches over the total area:

$$V_{run} = EIA\{ Depth - L_{int} - L_{max}(1 - exp^{(-Lrate\ (Depth - Lint))})\} \qquad \textit{[eq 3-1b]}$$

Figure 3.1 illustrates how rainfall depth relates to runoff volume for three ranges of (Lmax * Lrate). If this were greater than one, the runoff would briefly drop below zero. To prevent this, SIMPTM recalculates the rate to make the product equal one. Note also that only when it equals one is the rainfall-runoff curve smooth. For normal calibrations, vary either the rate or the maximum while keeping their product equal to one, unless observed data strongly indicates otherwise. Note that observed data often shows the rate more clearly.

FIGURE 3.1
VARIABLE RAINFALL LOSSES



The figure also shows the ease of calibrating EIA and Lmax. Even when pervious-area runoff is considered, its initial loss is so high (as much as one inch) that only impervious runoff is significant at lower depths. So look below one inch of rainfall to find the EIA and the initial loss.

Calibration often must rely upon data from other sites. Working with extensive data from Milwaukee, Wisconsin NURP and Toronto, Canada, Pitt (1987) showed that storms with rainfall less than 0.8 inch exhibit considerable losses even on impervious surfaces. These impervious losses varied by total rainfall depth and pavement texture. The constant initial loss (Lint) appeared much lower than previously accepted. Anything from 0.01 to 0.04 inch could be acceptable, depending on pavement texture. 0.02 inch has worked well in the Portland, Oregon area.

Recent calibration to 59 events in Surrey Downs and Lake Hills in Bellevue, Washington between April and September 1980 resulted in an Lmax of 0.05 inch and an Lrate of 20 per inch. Both basins had very smooth pavement. Pitt (1987) measured values for various surfaces:

Impervious Surface	Lmax	Lrate	Lmax*Lrate
Rough Streets	0.30	1.83	0.55
Smooth Streets	0.22	2.36	0.52
All Streets Combined	0.31	1.88	0.58
Large Paved Parking Areas	0.03	1.91	0.057
Large Flat Roof	0.08	1.02	0.082

An appendix describes how EIA can be estimated using mapped impervious area (MIA) determined from land use maps or aerial photos. Its simple equations depend upon the degree that impervious area is connected to the storm drain system. These were developed by Sutherland (1987) based on rainfall and runoff data collected by the USGS throughout the metropolitan areas of Portland and Salem, Oregon.

3.3 Impervious Sediment Washoff: Supply

SIMPTM uses published sediment transport equations to compute the capacity of simulated gutter or overland flow to remove various particle sizes of urban sediment. At each event time step and for each of the eight size groups, SIMPTM compares the current transport capacity of the flow to the remaining fraction of available sediment supply for each particle-size range. The lesser amount is transported.

Only a fraction of the accumulated sediment is initially available, and even more may become trapped by the armoring effects of loose bed material.

Dry Weather Accumulation

Research on urban storm water quality over the past two decades has identified particulate material from paved surfaces as the major source of pollutants found in urban runoff. The sediment accumulation rate on streets and other paved areas appears to decrease over accumulation time. Accumulation seems limited, regardless of the accumulation time. Data collected by many researchers suggest a first-order model with constant deposition (D) and resuspension by wind and traffic proportional to the accumulation by a first-order constant (K):

$$dP/dt = D - KP \qquad \text{[eq 3-1]}$$

Figure 3.2 illustrates the resulting exponential accumulation which relates the accumulation over time to the initial accumulation using the equilibrium Peq (D/K) and the resuspension rate (K):

$$P_{eq} - P(t) = (P_{eq} - P(0))\ exp^{-Kt} \qquad [eq\ 3\text{-}2]$$

Peq and K can be estimated from observed data by sampling street dirt over a period of time and analyzing dry period accumulations. While K might vary with particle size, insufficient data exist to quantify this. Alternatively, the data might indicate an equilibrium but the resuspension rate could be calibrated.

Without observed data, the accumulation parameters are probably the most difficult to estimate. Table 3.1 provides a summary of observed accumulation parameters found throughout North America. Note that its accumulation units are lbs. per curb mile, so when simulating overland flow, they must be re-expressed in lbs. per paved acre. In lieu of actual data collected on site, it is suggested that the Peq for paved acres be set at about one-third of the values shown. The K's shown represent a good starting point for the calibration process when stormwater quality data is available. However, limited experience has indicated that it can be at least two times greater. Note that in the table, the term Pmax means Peq.

Finally, though the initial accumulation is needed, it is an internal variable. So simulate a series of years and check it exactly one, two, or more years after the simulation start to find a good value. Often just looking one year later is sufficient.

The initial particle size distribution of sediment accumulated on paved areas and in catchbasin traps may be specified for the eight particle size groups. Considerable accumulation size distribution data has been collected throughout the United States and Canada since 1971. Extensive information exists for the following sites:

Location	Reference
Bellevue, Washington	Pitt 1985
Castro Valley, California	Pitt and Shawley 1982
Champaign, Illinois	Terstriep, Bender and Noel 1982
Ottawa, Canada	Pitt 1982
Reno/Sparks, Nevada	Pitt and Sutherland 1982
San Jose, California	Pitt 1979
Toronto, Canada	Pitt and McLean 1986

Table 3.2 summarizes the average distributions found in Reno/Sparks, Nevada. Figure 3.3 presents the average dry season distributions for Bellevue, Washington's Surrey Downs and Lake Hills and 148th Ave. sites. Table 3.3 summaries the distributions of both street dirt and parking lot dirt collected in May 1988 as part of the City of Portland's Columbia Slough Water Quality Management Program. The parameter input file key, Table 5-1, lists values used for all defaults which may be reset if desired in either the land-use or pollutant files.

Wet Weather Accumulation

Existing accumulation models work contrary to the accumulation patterns observed during many urban runoff events. While it is ***observed*** to be greater during times of large rainfalls (i.e. wet season), it is almost always ***modeled*** as being much less, because the rainfall washes accumulated street sediment into the drainage system. Seldom is any provision made for ***washon,*** the increase that is often observed to result when washoff from adjoining unpaved areas (e.g. gardens) actually increases the accumulation during an event. A few models require seasonal parameters to model accumulation differently during wet and dry seasons, but the season is arbitrary and seldom varies with rainfall depth. And furthermore, even during the wet season, the model is using the dry weather accumulation algorithms, albeit with larger values.

TABLE 3.1 STREET DIRT ACCUMULATION PARAMETER VALUES FOUND THROUGHOUT NORTH AMERICA

	Approximate Accumulation lbs/curb-mile				
	Initial Loading Value (PINT)	Maximum Observed Loading (PMAX)	Daily Deposition Rate (D) (lbs./curb mi./day)	Computed KVALUE (D/PMAX)	Sources
Smooth and Intermediate Textured Streets					
Reno/Sparks - good condition	280	300	4	0.013	1
Reno/Sparks - good condition with smooth gutters (windy)	880	1,400	25	0.018	1
San Jose - good condition	120	750	14	0.019	2
U.S. nationwide - residential streets, good condition	390	500	21	0.042	3
U.S. nationwide - commercial streets, good condition	300	500	14	0.028	3
Reno/Sparks - moderate to poor condition	710	710	7	0.010	1
Reno/Sparks - new residential areas	2,500	3,200	60	0.019	1
Reno/Sparks - fair to poor with lipped gutters	1,310	2,230	53	0.024	1
San Jose - fair to poor condition	280	810	14	0.017	2
Castro Valley - moderate condition	300	1,020	35	0.034	4
Toronto - moderate condition - residential	140	350	113	0.323	7
Toronto - moderate condition - industrial	210	1,240	141	0.114	7
Bellevue - dry period, moderate condition	500	800	25	0.031	6
Bellevue - heavy traffic	210	390	4	0.010	6
Bellevue - other residential sites	250	500	11	0.022	6
Champaign - all residential sites averaged	350	610	21	0.034	8
Average	546	957	35	0.047	
Range	120 - 250	300 - 3200	4 - 141	.01 - .323	
Rough and Very Rough Textured Streets					
San Jose - oil and screens overlay	1800	2600	21	0.008	2
Reno/Sparks	2,230	3,040	35	0.012	1
Reno/Sparks - windy	1,910	5,000	120	0.024	1
San Jose - Poor condition	780	1,520	21	0.014	2
U.S. nationwide - industrial streets, poor condition	670	1310	35	0.027	3
Average	1,478	2,694	46	0.017	
Range	670 - 2230	1310 - 5000	21 - 120	.008 - .027	

Sources:

(1) Pitt and Sutherland 1982
(2) Pitt 1979
(3) Sartor and Boyd 1972 (corrected)
(4) Pitt and Shawley 1982
(5) Pitt 1982
(6) Pitt 1985
(7) Pitt and McLean 1986
(8) Terstriep, Benderand Noel 1982

TABLE 3.2

AVERAGE PARTICLE SIZE DISTRIBUTIONS OF STREET DIRT FOUND IN RENO/SPARKS NEVADA

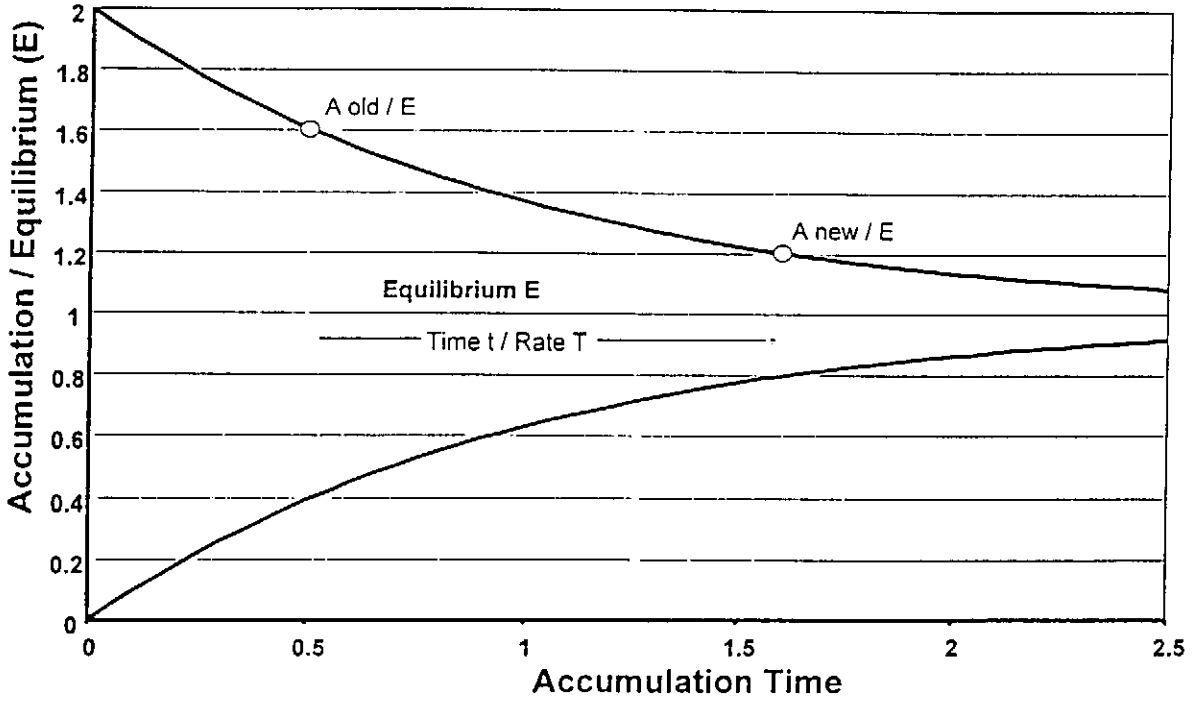
Sampling Sites	Particle Size Groups (microns)							
	< 63	63–125	125–250	250–500	500–1000	1000–2000	2000–6370	> 6370
Mill	0.07	0.12	0.17	0.22	0.20	0.12	0.09	0.02
Tallman	0.10	0.11	0.16	0.29	0.22	0.09	0.03	0.01
Trainer	0.14	0.18	0.19	0.17	0.14	0.10	0.06	0.03
Locust	0.07	0.13	0.22	0.21	0.15	0.10	0.08	0.05
Brookfield	0.10	0.11	0.16	0.21	0.18	0.14	0.09	0.02
CBD	0.07	0.13	0.19	0.22	0.17	0.10	0.08	0.05
Parkview	0.09	0.14	0.20	0.19	0.16	0.11	0.08	0.04
Glendale	0.07	0.09	0.14	0.18	0.21	0.17	0.12	0.04
Lepori	0.08	0.07	0.12	0.17	0.23	0.21	0.12	0.01
Fifth	0.07	0.14	0.24	0.22	0.16	0.08	0.06	0.02
Smooth site averages (1)	0.08	0.13	0.20	0.23	0.18	0.09	0.06	0.03
Intermediate texture site averages (2)	0.10	0.15	0.21	0.19	0.15	0.10	0.07	0.04
Rough texture site averages (3)	0.08	0.10	0.15	0.19	0.21	0.16	0.11	0.02

(1) Smooth sites: CBD, Fifth, Tallman, and Parkview
(2) Intermediate texture sites: Trainer and Locust
(3) Rough texture sites: Lepori, Glendale, Mill, and Brookfield

Source: Pitt and Sutherland 1982

FIGURE 3.2

Equilibrium Accumulation Function



$$A_{new} = (A - E)(e^{-t/T} - 1)$$

FIGURE 3.3

Dry Season Particle Size Distributions
Found in Bellevue, WA

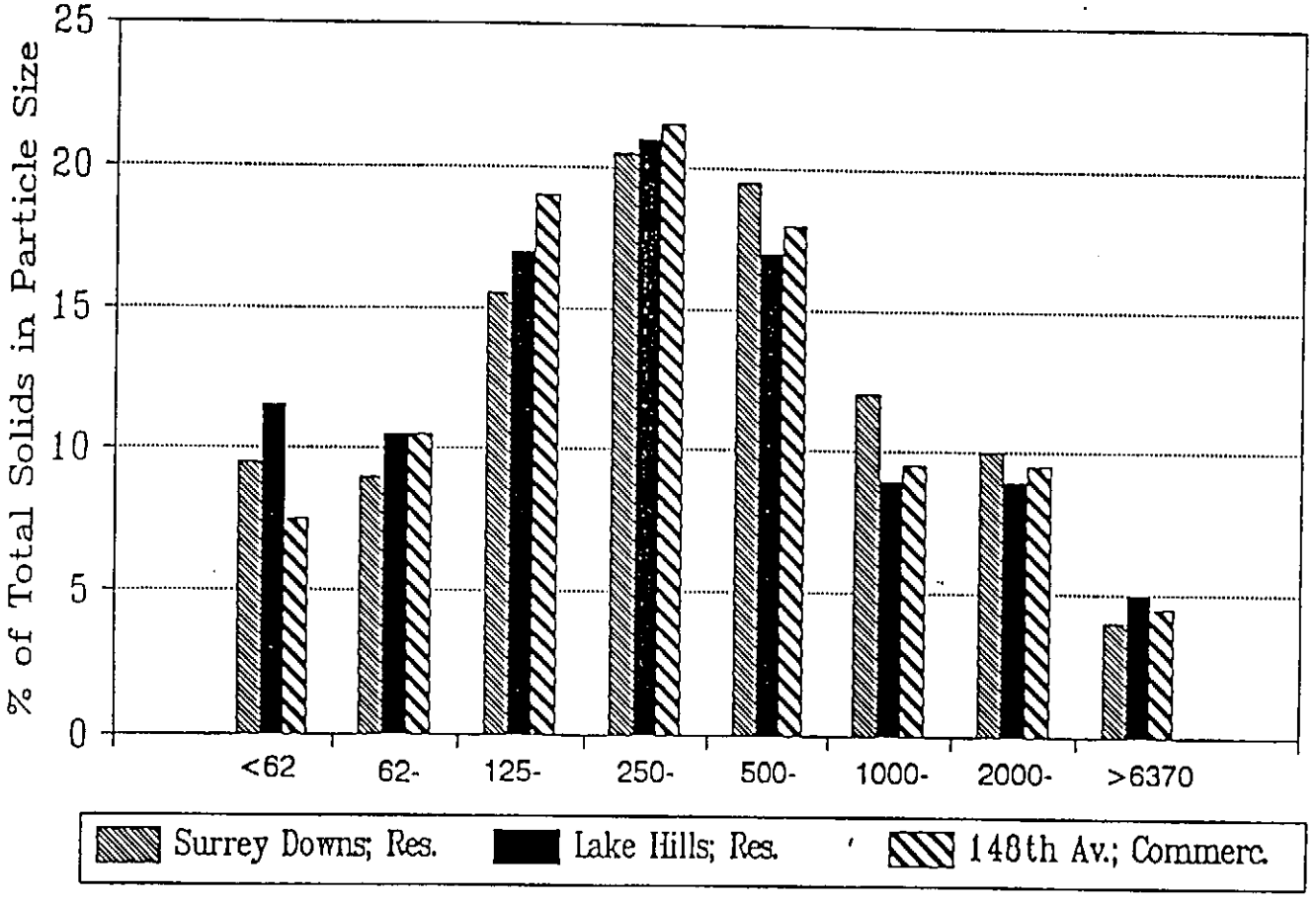


TABLE 3.3

PARTICLE SIZE DISTRIBUTION AND LOADINGS FOR COMPOSITE SAMPLES OF PAVED SURFACE PARTICULATES LOCATED THROUGHOUT TRIBUTARY AREAS OF THE COLUMBIA SLOUGH

		PARTICLE SIZE DISTRIBUTION (MICRONS) AND LOADINGS FOR COMPOSITE SAMPLES							
IDENTIFICATION		< 38	38–63	64–125	126–250	251–417	418–1981	> 1981	TOTAL
INDUSTRIAL	lb/curb mi	8.6	15.8	38.1	75.2	114.3	219.4	36.6	508
	lb/paved ac	3.4	6.2	14.9	29.5	44.8	85.9	14.3	199
	percent	1.7	3.1	7.5	14.8	22.5	43.2	7.2	100
INDUSTRIAL TRACK OUT	lb/curb mi	53.1	53.1	99.6	208.1	229.2	324.4	139.5	1107
	lb/paved ac	20.8	20.8	39.0	81.6	89.9	127.2	54.7	434
	percent	4.8	4.8	9.0	18.8	20.7	29.3	12.6	100
SINGLE FAMILY RESIDENTIAL (under construction)	lb/curb mi	216.4	296.4	569.1	764.7	243.0	643.2	231.2	2964
	lb/paved ac	85.0	116.4	223.5	300.3	95.4	252.6	90.8	1164
	percent	7.3	10.0	19.2	25.8	8.2	21.7	7.8	100
SINGLE FAMILY RESIDENTIAL (steep sloped streets)	lb/curb mi	0.8	1.5	7.1	19.6	33.1	161.6	152.3	376
	lb/paved ac	0.4	0.8	3.8	10.4	17.5	85.5	80.6	199
	percent	0.2	0.4	1.9	5.2	8.8	43.0	40.5	100
SINGLE FAMILY RESIDENTIAL (flat sloped streets)	lb/curb mi	1.9	4.9	12.7	21.0	25.4	92.7	53.4	212
	lb/paved ac	1.1	2.9	7.4	12.3	14.9	54.2	31.2	124
	percent	0.9	2.3	6.0	9.9	12.0	43.7	25.2	100
MULTI-FAMILY RESIDENTIAL	lb/curb mi	6.8	10.8	21.2	41.5	74.0	206.5	90.2	451
	lb/paved ac	4.0	6.4	12.5	24.5	43.6	121.8	53.2	266
	percent	1.5	2.4	4.7	9.2	16.4	45.8	20.0	100
COMMERCIAL STREETS	lb/curb mi	14.0	28.7	72.7	111.6	115.5	171.7	123.8	638
	lb/paved ac	6.5	13.3	33.7	51.8	53.6	79.7	57.4	296
	percent	2.2	4.5	11.4	17.5	18.1	26.9	19.4	100
COMMERCIAL PARKING LOTS	lb/curb mi	5.2	9.5	18.5	27.1	36.6	97.6	113.5	308
	lb/paved ac	2.6	4.7	9.1	13.4	18.1	48.2	56.0	152
	percent	1.7	3.1	6.0	8.8	11.9	31.7	36.9	100

Source: OTAK, Inc. 1989

SIMPTM has changed all that. SIMPTM uses a modified exponential washon curve to add extra sediment to that accumulated through atmospheric (dry) deposition only. This washon will not approach zero as does atmospheric (dry), but rather will approach a constant rate (proportional to additional rainfall) at larger rainfalls. In this way it is no longer necessary to hassle with a separate pervious runoff that is generally very difficult to isolate from observed rainfall-runoff data.

The following 2-parameter model is used to represent wet weather accumulation that occurs during rainfall events when rainfall depth (Drun) exceeds a minimum threshold depth (Dmin):

$$WetAccum = Cmax *(Drun - Dmin *(1 - exp(- Drun / Dmin))) \qquad [eq\ 3\text{-}2a]$$
$$WetAccum = 0 \text{ if } Drun < Dmin$$

Figure 3.3a illustrates the behavior and identifies the parameters associated with the wet weather accumulation model.

With this model, the parameter Cmax is specified in terms of units of accumulation (which may vary depending upon whether overland flow or curb-and-gutter drainage is simulated. Cmax is also referred to as WetAccS in Section 4 File Formats. Thus this option may be easily activated with a non-zero Cmax, as when Cmax is zero wet accumulation is also zero.

Dmin, the threshold depth to trigger wet accumulation, is specified in inches of rainfall. Dmax is referred to under the section on input formats as WetAccX. Note that the X is for the X axis of a graph of wet accumulation versus rainfall depth.

Availability

Only a fraction of the total accumulated sediment or total solids is available for washoff. Pitt (1987) in a series of simulated rainfall tests concluded that it ranged from 0.028 to 0.26. The lowest value occurred on rough textured, dirty streets with a low rainfall intensity of 0.11 inches per hour, while the highest occurred on smooth textured, cleaner streets with an intensity of 0.48 inches per hour.

SIMPTM correlates the available fraction to runoff volume up to a maximum fraction (SFmax) at the runoff depth in inches (SFrun):

$$Sedfrac = SF_{max}\ (Rundep / SF_{run})2$$

Recent calibrations to Bellevue, Washington NURP data resulted in values of 0.15 and 0.04 for the SFmax and SFrun, respectively.

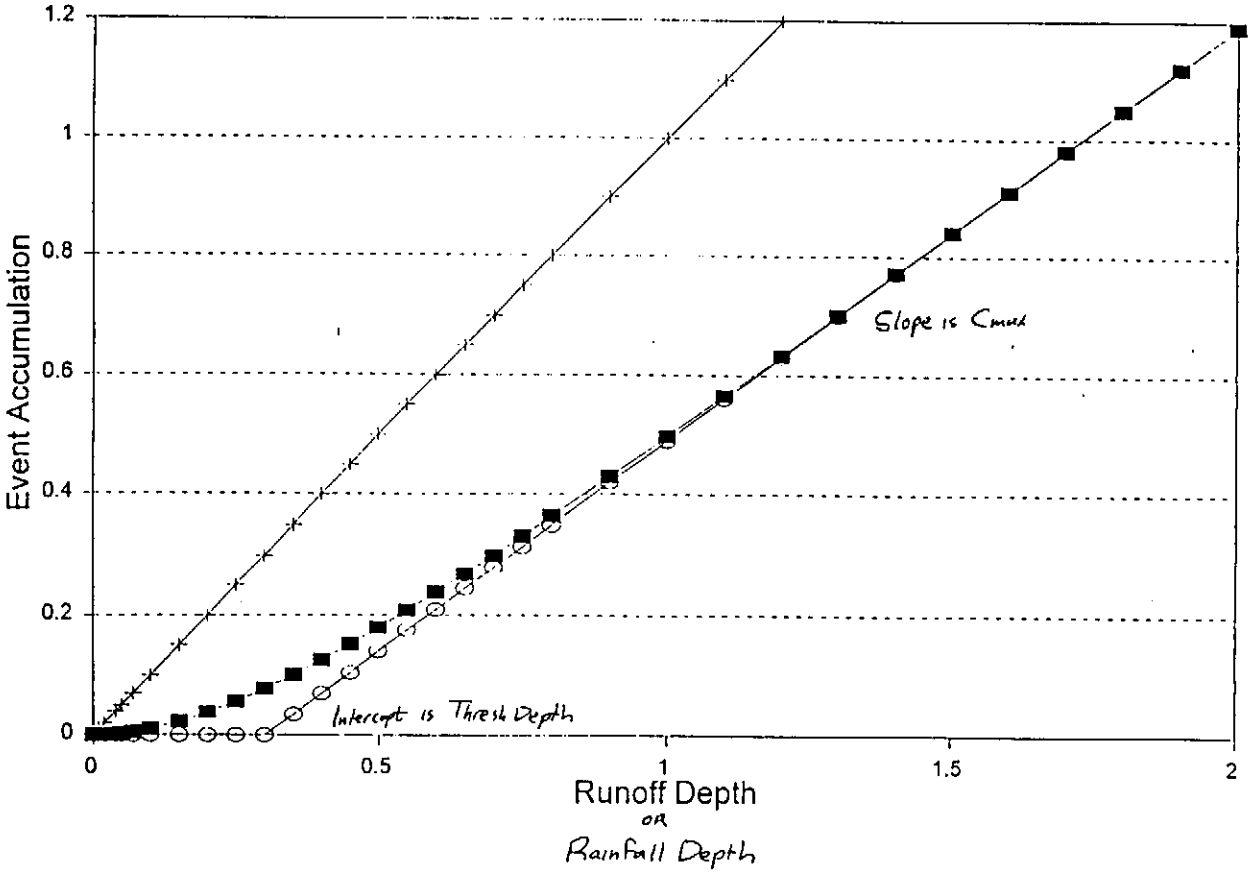
Bed Armoring

Bed armoring further limits the available supply of sediment when large stable particles shelter smaller, otherwise unstable particles. Research indicates that it depends upon the composition of the armored bed and the amount of particles already eroded (Vanoni, 1975). SIMPTM constantly checks the stability of the largest particle size group with available sediment. When this size group is stable, bed armoring is considered. When only 5% of a group's accumulation remains during an event, the group can no longer shelter smaller-sized groups.

SIMPTM monitors armoring for each affected size range separately. When armoring accurs at the start of an event, the initially available sediment (Po)is weighted by the proportion of the total bed weight (Ptot) to find the resulting fraction available:

$$P_{avail}/P_o = ½\ P_o / P_{tot}$$

FIGURE 3.3A
Wet Weather Accumulation



Throughout the remainder of the event, the fraction available in each group relates the remaining available sediment (P) and the initial available sediment:

$$P_{avail}/P = 1 - 1P/P_o$$

Thus for a given particle size range, the fraction of sediment currently available for transport will increase as more of its sediment is eroded, an observation by Gessler (Vanoni, 1975).

Street Cleaning

SIMPTM can simulate street particulate removals by scheduled street cleaning. To invoke the street sweeping option, specify the scheduled number of days between cleanings and the year, month, day and hour of the first cleaning operation. In addition, Park - the fraction of curb blocked by parked cars and thus not swept — must be specified. This effect can be ignored by specifying zero. If sweeping is scheduled during an event, SIMPTM defers it until the next scheduled time, as do most public works departments.

Studies by Sartor and Boyd (1972) and Pitt (1979), demonstrated that street cleaning efficiency depends upon the particle size distribution of the accumulated material. Street cleaning equipment usually picks up larger particles better. Additional studies by Pitt and Shawley (1982) and Pitt and Sutherland (1982) found it to also depend upon the amount of accumulation: sweeping removes little, if any, material below a certain base residual, which varies by particle size.

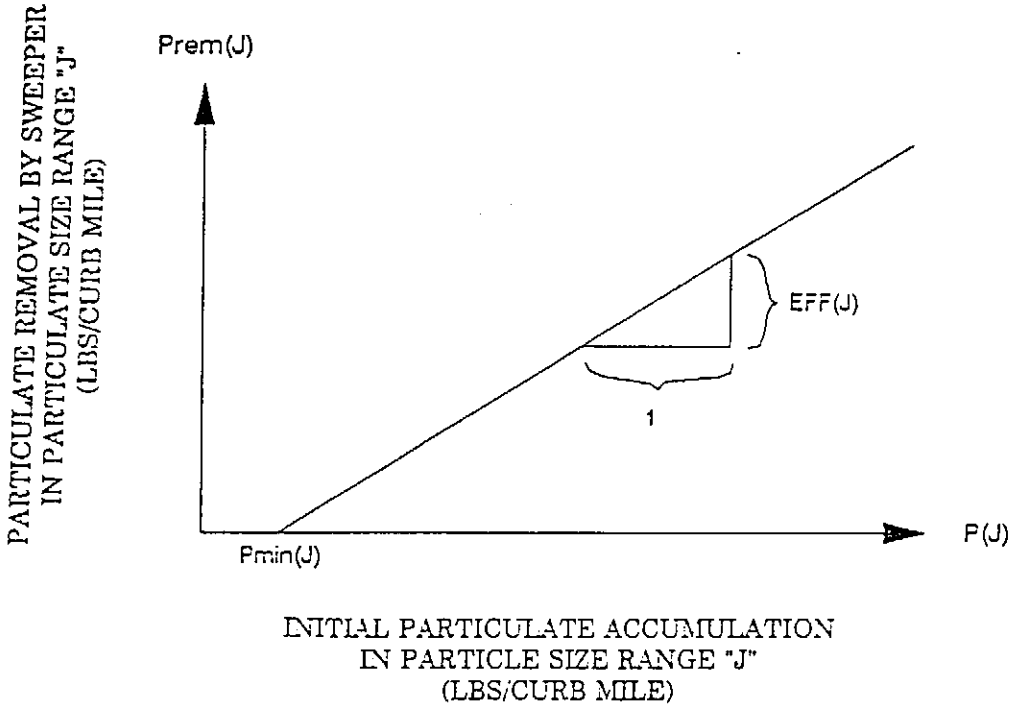
Figure 3.4 illustrates the street cleaning component and equations used by SIMPTM. For each size group, the amount removed (Prem) is proportional to the accumulation (P) in excess of the base residual (SSmin) by a sweeping efficiency (SSeff):

$$P_{rem} = SS\ eff(P - SS_{min})\ for\ P > SS_{min} \qquad [eq.\ 3\text{-}3]$$

For convenience, defaults were encoded; they are listed in Table 4-1. However, the type of cleaning equipment used can affect these numbers. These 16 parameters can be estimated from street cleaner productivity data analyses, often provided by the manufacturer. These parameters have been measured in several locations:

Location	Reference
Bellevue, Washington	Pitt 1985
Castro Valley, California	Pitt and Shawley 1982
Chapaign, Illinois	Terstriep, Bender and Noel 1982
Reno/Sparks, Nevada	Pitt and Sutherland 1982
San Jose, California	Pitt 1979

FIGURE 3.2
SIMPTM's Street Cleaning Component



Pmin(J)= THE BASE RESIDUAL LOADING OF PARTICULATE SIZE RANGE "J" (LBS/CURB MILE)

EFF(J)= THE STREET CLEANING EFFECTIVENESS AS A FRACTION OF PARTICULATE LOADINGS IN EXCESS OF THE BASE RESIDUAL FOR SIZE RANGE "J"

The results varied considerably but regenerative air cleaners were usually more effective with smaller particles and less so with larger ones than were traditional mechanical cleaners. The following ranges are reasonable:

Particle Size Group Number	Range of Particle Size in Microns	SSmin (lbs./curb mile)	SSeff
1	less than 62	5-25	0.35-0.80
2	62-125	15-30	0.40-0.85
3	126-250	20-35	0.40-0.85
4	251-500	20-35	0.40-0.85
5	501-1000	10-25	0.30-0.80
6	1001-2000	5-20	0.25-0.75
7	2001-6370	5-15	0.20-0.85
8	greater than 6370	0-10	0.20-0.85

In fact, the regenerative air cleaners inability to pick up the larger visible material including litter and other debris has overshadowed its superior water quality related benefits. Some larger cities have been successfully using regenerative air cleaners as the second cleaner in a tandum cleaning operation in which the mechanical cleaner sweeps first.

Several Nationwide Urban Runoff Program (NURP) studies concluded that street cleaning was ineffective in significantly reducing the event mean concentrations of most pollutants found in urban runoff. Some of these studies indicated that the event mean concentrations even appeared to be have increased. Working with an earlier version of the SIMPTM program Sutherland (1980) reported that model simulations on a hypothetical subcatchment whose slope was greater than 1.5% seem to suggest that intensive street cleaning operations may increase event mean concentrations of pollutants. It was concluded at that time that cleaning operations removed a greater portion of the larger particles which if not removed would have pinned or armor a larger portion of the smaller more highly polluted particles. Once these larger particles are removed by street cleaning, more smaller particles become available and are subsequently transported by the lower intensity storms that occur more frequently.

The paper also concluded that cleaning streets more frequently than once every two weeks in tandem with on-site retention or regional detention would provide dramatically higher removals than just the retention or detention controls alone. This conclusion is based on the belief that these more frequent lower intensity and volume storms were delivering greater quantities of pollutants to the retention or detention facility which in turn provided a much greater performance for these smaller volume storms. Unfortunately many water quality planners are probably overlooking the real water quality related potential for street cleaning operations which is to clean those areas that are already served by a well designed retention or detention facility.

3.4 Impervious Sediment Washoff: Capacity

SIMPTM divides transport capacity into two components: the bed load and suspended load capacities. The bed load sediment transport capacity is based on the Yalin equation (Yalin, 1963) as modified and experimentally verified by Foster and Meyer (1972). The suspended load capacity is based upon Einstein (1950).

Some parameters, such as the gravity constant (g, 32 ft/sec2), sediment specific gravity (SG, 2.65) and the viscosity of water (v, 1.41E-5 ft2/sec) are coded into the model. Most of the others required can be obtained from field reconnaissance, topographical maps, street maps, roadway design standards and land use maps. These include CBDN, CurbDn, Slope, Z (Cross slope for Curb and Gutters) and Nvalue. The Manning's roughness coefficient is based on shallow flow hydraulics and thus is a factor of ten higher than those

appropriate for flood hydraulics. Try 0.08 and 0.12 for curb-and-gutter and overland-flow systems, respectively.

One washoff parameter that requires some attention is WIDTH, the representative width of the overland flow contributing to a stormwater inlet. This parameter is only used in overland flow, where it greatly influences the representative flow depth and sediment transport capacity. It shares the same input position with the cross slope for curb-and-gutter systems. In lieu of actual on-site data, one technique to estimate it might be to determine the minimum value that provides some washoff for the lowest volume and intensity event having significant runoff. Then determine how it affects the washoff of an average event within the continuous trace. One must then assess the overall validity of the mean event concentrations simulated. It is anticipated that WIDTH could easily vary from 20 to 100 feet depending on the physical characteristics of the areas tributary to the average inlet. 60 feet has typically been used for Pacific Northwest parking lots.

Bed Load

The capacity of shallow flow to move bed sediment is based largely upon the local hydraulic conditions and the particle size and specific gravity. At each time step during the simulation, SIMPTM divides the outlet flow from the trapezoidal hydrograph by the number of storm water inlets in the subcatchment. This is then halved to obtain the representative flow and hydraulic radius (r_H) at the simulated representative point, from which eight Reynolds numbers are obtained for the eight group particle diameters (DIA):

$$N_{re} = V^* \, DIA / \nu \qquad\qquad V^* = (g \, r_H \, Slope)^{1/2}$$

Shields Diagram, illustrated in Figure 3.5, relates these numbers to a dimensionless critical shear (σ_{cr}), which provides a maximum particle size (DIAmax) that can be transported for each group:

$$\sigma_{cr} = Slope * r_H/DIA_{max} * 1/(SG - 1)$$

If a group size is smaller than its critical diameter, the group dimensionless excess shear becomes non-zero:

$$\sigma = DIA_{cr}/DIA - 1$$

The cited work experimentally related the dimensionless excess shear, critical shear, and specific gravity to a dimensionless potential of entrainment for each group, were it the only group present:

$$PE_{alone} = 0.635 \, \sigma \, (1 - Ln(1 + \Delta))/\Delta \qquad\qquad \Delta = 2.45 \, SG^{-0.4} \, \sigma_{cr}^{1/2} \, \sigma$$

This number had been defined when the bed transport capacity (CAPB) was non-dimensionalized:

$$CAP_{alone} = (\rho g \, DIA \, V^*) * PE_{alone}$$

However, the bed is not of a single-size, but is modeled as having eight size groups of particles. Were these single-size capacities just added, the fluid would be "filling" its full capacity on EACH size group. So these eight single-size-bed capacities are weighed by each excess shear's proportion to their total (σtot) to obtain the actual amount transported for each group:

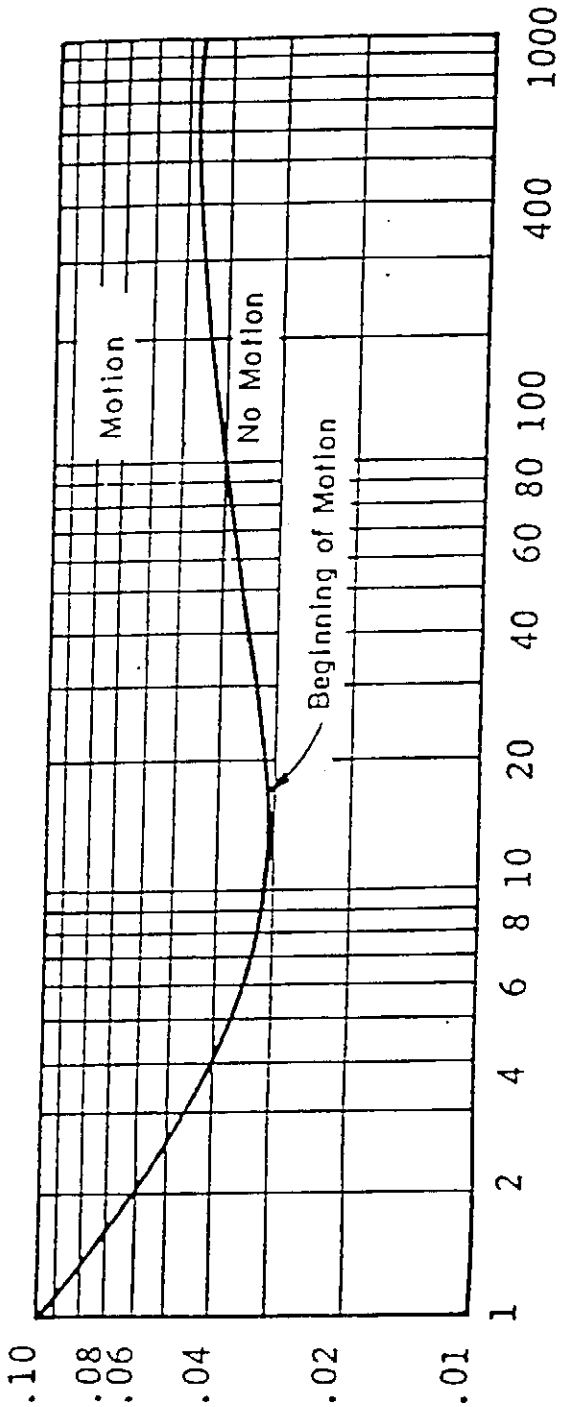
$$CAP_{bed} = CAP_{alone} * \sigma / \sigma_{tot}$$

Suspended Load

A group's suspended load begins where its bed load ends: twice the particle diameter (DIA). Einstein (1950) found the suspended load proportional to the bed load:

$$CAP_{suspended} = CAP_{bed} \{E^{w-1} / 11.6 \, (1-E)^w\} * \{J1 \, V/V^* + 2.5(J1 + J2)\}$$

FIGURE 3.5
Shields Diagram of Sediment Grain Stability



Form Drag vs. External Bed Forces:
σ = *Slope* $(r_H / DIA) / (SG - 1)$

Form Drag vs. Viscous Drag Forces: $N_{re} = V^* DIA / \nu$

Shear Velocity: $V^* = (g\ r_H\ Slope)^{1/2}$

Adapted from Richardson (1990) and Gessler (1971)

J1 and J2 are integrals resulting from the integration of the vertical concentration of sediment in the flow and the vertical velocity distribution with average V. They are solved using an algorith developed and verified at Colorado State University (Li, 1974; Simons, Li and Ward, 1977). The researchers applied the work of Einstein (1950) to a shallow flow problem. After non-dimensionalizing the problem, the range of integration transforms from between the bed boundary and the flow depth (y) to between E (2DIA/y) and 1. The dimensionless ratio of settling velocity to shear velocity is also used:

$$w = V_s / (0.4\ V^*)$$

The settling velocity (Vs) varies by particle size and is computed for each group:

$$V_s = 2.9517\ DIA^2 / v \qquad DIA < 60\ \mu m$$
$$V_s = \{(36.064\ DIA^3 + 36v^2)^{1/2} - 6v\} / DIA \qquad DIA >= 60\ \mu m$$

3.5 Pervious Sediment Washoff

Although pervious surfaces usually contribute little to washoff during typical events, large enough storms can fill the depression storage separating these areas from the drainage system and allow pervious washoff to noticeably contribute to the resulting pollutant loadings. Look outside during an intense rainfall event: fans of soil and debris radiate onto paved streets or parking lots from small adjoining pervious areas which would not pool enough to overflow during smaller events. However, these large deposits often don't completely washoff but contribute instead to the impervious area deposition rate. SIMPTM simulates sediment washoff from pervious areas at a constantly decreasing concentration with pervious runoff depth (Runperv) up a maximum load per pervious acre (Lperv) at a pervious runoff depth (Dperv):

$$Pervwash = L_{perv}\,(1 - (1 - Run_{perv} / D_{perv})^2)$$

SIMPTM simulates pervious and impervious runoff alike, discussed in the runoff section. However, to reflect the significant initial storage and losses, the constant initial loss Lint will be much higher for the pervious areas. Also, the variable loss must reflect greater infiltration, depression storage, and the amount of pervious area that never contributes. The product of Lmax*Lrate must certainly be one, as the pervious runoff increases must increase gradually but smoothly. Pitt (1989) measured runoff coefficients which increased with rainfall depth for several area types, including mixed C and D soils from Milwaukee, Wisconsin, and B soils in Toronto, Canada. This data was recently fit to the variable loss parameters used:

Pervious Surface	Lmax	Lrate	Lmax*Lrate
Type B Toronto Soils	7.7	0.13	1.0
Type C-D Milwaukee Soils Mixed	4	0.25	1.0

Recent calibration to Bellevue, Washington data indicates that even 1 inch would be an appropriate initial pervious loss, for that seemed to be when the pervious areas began to contribute. Although Pitt measured initial losses of 0.2 and 0.12 respectively, when only pervious areas were tested. In fact, the high initial losses calibrated (and easily observed by rainy-day field surveys) often arise at the interface between the pervious plots and the paved areas. Calibrated variable losses seemed reasonable.

3.6 Sediment Trapping Catchbasins

To invoke the sediment trapping option, specify the average base-area, total trap depth and initial depth of sediment stored. If the catchbasins are to be cleaned, specify a positive number of days between scheduled cleanings and the year, month, and day of the first cleaning. Otherwise, these must be zero.

Very little research has been conducted on sedimentation in stormwater inlets or catchbasins. However, experiments by Lager et al. (1977) concluded that sediment accumulation in catchbasins is a function of the incoming sediment sizes, the catchbasin volume available to trap sediment, and the runoff flow rate entering the catchbasin. As expected, the larger particles had the highest retention percentage.

An initially clean catchbasin would retain approximately 45 percent of the incoming sediment until its became about half full. The efficiency then quickly drops to 0 as the trap fills to 60%. This critical point was defined as the breakthrough point. SIMPTM uses log-log regressions to relate the capture rate for each group to the retention rate (X), or flow over available trap storage exponentially:

$$Capfrac = \tfrac{1}{2}(1 - A_{trap} X^{(Btrap)})$$

The two parameter sets Atrap and Btrap are coded into the program. Each set of eight size-group values can be re-defined.

3.7 Additional Pollutants

SIMPTM associates each pollutant other than total sediment or solids to total solids using eight pollutant-to-sediment mass fractions or pollutant strengths for each of the eight sediment size groups. For convenience this information may be in either the Land-Use or Pollutant Files. If many land uses are required for basin-wide planning efforts, using a pollutant file for pollutant strengths and control strategies centralizes information and promotes better quality management. Simple, single-land use simulations can still use strengths in their Land Use File. Although different strengths in different files may be used for different contributing land uses, ensure that the pollutant names and order remain consistent throughout all related files. Fortunately, SIMPTM does ignore case and any leading or trailing blanks when checking.

Pollutant strengths can be determined by sampling accumulation on paved surfaces or catchbasin traps, separating into eight constituent size groups and testing for the mass fraction of each desired pollutant present. However, collecting enough data to calibrate to can be costly. Fortunately, they can be calibrated to pollutant loading or event mean concentration data once total solids are calibrated. The pollutant strengths can be adjusted to match the observed washoffs from increasingly intense storms, from those with just groups 1 and 2 washing off to those with all eight. Considerable care is needed to assess the results of this effort in relation to previously measured data.

Considerable pollutant strength data has been collected throughout the United States and Canada, and is available for all locations listed earlier as having particle size distribution data:

Location	Reference
Bellevue, Washington	Pitt 1985
Castro Valley, California	Pitt and Shawley 1982
Champaign, Illinois	Terstriep, Bender and Noel 1982
Ottawa, Canada	Pitt 1982
Reno/Sparks, Nevada	Pitt and Sutherland 1982
San Jose, California	Pitt 1979
Toronto, Canada	Pitt and McLean 1986

Table 3.4 summarizes the pollutant strengths obtained from the Bellevue, Washington Nationwide Urban Runoff Program (NURP). Each represents an average of about ten separate samples collected from two residential drainage basins between March 1980 and January 1982. Table 3.5 reports the limited pollutant strengths sampled in May 1988 for the City of Portland's Columbia Slough Water Quality Management Program.

TABLE 3.4

AVERAGE VALUES OF STREET DIRT POLLUTANT STRENGTHS (mg/kg)
FOUND IN BELLEVUE WASHINGTON

Pollutant	Basin	Particle Size (Microns) < 63	63–125	126–250	251–500	501–1000	1001–2000	2001–6350	> 6350
Chemical Oxygen Demand (COD)	Surrey Downs – Main	180,000	150,000	100,000	94,000	130,000	190,000	170,000	280,000
	Surrey Downs – WHR *	190,000	190,000	150,000	150,000	170,000	280,000	300,000	380,000
	Surrey Downs – 108th *	150,000	100,000	54,000	36,000	37,000	37,000	55,000	70,000
	Surrey Downs – Average	173,333	146,667	101,333	93,333	112,333	169,000	175,000	243,333
	Lake Hills	230,000	180,000	110,000	100,000	210,000	240,000	270,000	420,000
	Overall Average	187,500	155,000	103,500	95,000	136,750	186,750	198,750	287,500
Total Kjeldahl Nitrogen (TKN)	Surrey Downs – Main	2,900	2,600	1,700	1,300	1,400	1,600	1,200	1,500
	Surrey Downs – WHR *	3,300	3,300	2,300	1,700	1,900	2,500	2,400	1,800
	Surrey Downs – 108th *	1,800	1,200	970	410	460	340	290	350
	Surrey Downs – Average	2,667	2,367	1,657	1,137	1,253	1,480	1,297	1,217
	Lake Hills	3,500	3,200	1,900	1,600	2,300	2,100	2,000	3,200
	Overall Average	2,875	2,575	1,718	1,253	1,515	1,635	1,473	1,713
Total Phosphorus (TP)	Surrey Downs – Main	830	610	470	420	480	690	750	740
	Surrey Downs – WHR *	810	680	470	520	530	580	640	620
	Surrey Downs – 108th *	690	510	330	300	380	620	640	620
	Surrey Downs – Average	777	600	423	413	463	630	677	660
	Lake Hills	940	740	550	440	570	760	740	750
	Overall Average	818	635	455	420	490	663	693	683
Total Lead (TPb)	Surrey Downs – Main	1,400	1,200	1,100	840	680	420	240	280
	Surrey Downs – WHR *	440	330	250	180	160	320	50	80
	Surrey Downs – 108th *	1,600	1,400	1,400	910	570	240	130	90
	Surrey Downs – Average	1,147	977	917	643	470	327	140	150
	Lake Hills	1,900	1,900	1,700	1,300	900	630	350	210
	Overall Average	1,335	1,208	1,113	808	578	403	193	165
Total Zinc (TZn)	Surrey Downs – Main	320	260	210	170	160	120	110	100
	Surrey Downs – WHR *	180	140	100	80	75	100	75	85
	Surrey Downs – 108th *	270	210	160	120	130	130	100	150
	Surrey Downs – Average	257	203	157	123	122	117	95	112
	Lake Hills	370	330	270	220	180	180	130	140
	Overall Average	285	235	185	148	136	133	104	119

* Both Surrey Downs – Westwood Homes Road (WHR) and 108th Avenue S.E. (108th) are uncurbed streets.

Source: Pitt, 1985

TABLE 3.5

AVERAGE PAVED SURFACE PARTICLE POLLUTANT STRENGTHS FOUND IN URBAN AREAS TRIBUTARY TO THE UPPER COLUMBIA SLOUGH

Area Description	Particle Size Range (microns)	Pollutant Strength (mg/kg)				
		COD	Total Kjeldahl Nitrogen	Total Phosphorus	Lead	Zinc
Commercial	< 63	58600	969	22	635	365
Land	63–250	43200	541	44	760	317
Use	> 250	47000	1153	11	286	278
Industrial	< 63	57800	1589	46	153	174
Land	63–250	25400	748	47	136	152
Use	> 250	45700	620	7	63	122
Residential	< 63	64800	1600	39	207	315
Land	63–250	48900	1120	41	345	195
Use	> 250	45100	656	13	134	130
Industrial Track Out*	< 63	22200	1010	48	77	137
Residential Under Const.**	< 63	12100	387	59	15	54

* Industrial Track Out is a light industrial area whose street surface loadings were influenced by sediment tracked out by vehicles leaving an adjacent construction site.

** Residential Under Construction is a single family residential area whose street surface loadings were influenced by sediment transport by runoff flowing from a construction site.

Source: OTAK, Inc. 1989

SECTION 4

FILE FORMATS

Contents at a glance

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Note – inserted 11x17 page of input variables

4

SECTION 4
FILE FORMATS

The parameter input files are all vital to the SIMPTM package success. Care has been taken to ensure that they are easy to use and can relate clearly with finished tables of land use category parameters such as in Table 4.2. All input files are FREE FORMAT except for hourly rainfall data. This means that there is no more worry over column offsets. However, all parameters must be given a value. Those not used will be zero. Finally, the optional pollutant strengths and default resets may appear both in the Pollutant File and the Land-Use File, but input information is listed only for the Pollutant File for better clarity.

In the line-by-line descriptions, parameter names precede values. In some cases, specific range limits are listed; in others, "+" or "—", indicating sign, or specific values select simulation options. "A" values indicate text. Input lines should be no longer than 120 characters. Fold out Table 5.1 to see the file layouts while reviewing the line-by-line requirements below:

4.1 RAINEV Instruction File (*.IN)

The instruction file determines how the raw hourly rainfall data is divided into events. After event selection parameters have been defined, only one run of RAINEV is needed. Afterwards, the event log file can be re-read and re-summarized by month, parameter, or year. Finally, ALL rainfall depths for RAINEV are in HUNDREDTHS of an INCH. *One key point: title lines must NOT begin with numbers.*

The first line determines the arrangement of subsequent input, selects the raw hourly precipitation data format, selects the summary table and indicates whether an event log file is to be written. Note that at least one output must be specified. If the RN file is read, a summary must be output.

Ntitle	Simptm	Output	Input

Name	Value	Description
Ntitle	+	Number of title lines to follow, including Project and Rntitle (**IGNORED**)
Simptm	+	Events are fit to trapezoids; event log (*.RN) is written (for SIMPTM)
	0	Event Log File not requested
Output	+	Output summaries (see below for forms) are spreadsheet ready
	—	Finished summaries where spreadsheets are not available
	±3	Event statistics are summarized for each YEAR by month and parameter
	±2	Event statistics are summarized for each PARAMETER by year and month
	±1	Event statistics are summarized for each MONTH by year and parameter
	0	Event statistics are not summarized
Input	1	Alternative 1-line, 1-hour rainfall format; hours from 1 to 24: Year, Julian Day, Hour, Rain (Int, hundredths of inch); *free format*
	2	Flexible/Free CD-ROM Format; hours from 1 to 24 (25 is skipped daily total): Month, Day, Year, Hour, Rain (real, inches); *free format*
	3	Fixed CD-ROM Format; hours from 1 to 24 (25 is skipped daily total): "STID|PO|MMDDYYYY|HHMM|rain|note" (real rain, inches; any char for "|")

The first title line is the Project Title which identifies all subsequent files. Later titles which differ from this will produce a warning message on the screen. Only the first 100 characters will be checked. Note that this file MUST have at least one title, and that the project title is included in the total number of titles above.

The second line, if present, describes the rainfall record used. It will be combined with a brief list of event parameters used and retained throughout the later project files. The first 60 characters must correspond throughout the project, and identical event parameters must have been used. If the second title is absent, the first title (above) will be used instead. This title also must be counted in the total number of titles above.

Any number of additional titles desired may follow, but will not appear in any output. All must be included in the number of title lines.

The final line contains the event location parameters discussed in the RAINEV description.

Lapse	Trace	Thresh1	Thresh3	Thresh6

Name	Value	Description
Lapse	+	Consecutive dry hours separating events
	—,0	Lapse defaults to 3 hours
Trace	0+	Minimal hourly rainfall; rainfall below this is ignored
Thresh1	+	Minimum 1-hour depth required to recognize an event
Thresh3	+	Minimum 3-hour depth required to recognize an event
Thresh6	+	Minimum 6-hour depth required to recognize an event

in hundreds of an inch

4.2 SIMPTM Land Use File (*.LU)

Each Land Use File describes a land-use category. If other pollutants are to be summarized, the pollutant strengths may either be included here or in a Pollutant File. Similarly, control strategies may be invoked and defaults be reset in either file. In both cases, this Land Use File takes precedence, and only the last of any type of action is remembered. These option lines are commented out if any of ";","!", or "*" are present anywhere on the line.

The first line determines the arrangement of the remaining file, and selects the output options. All outputs are spreadsheet-ready "numbers" or "quote-delimited" files. The listing of event may optionally include event mean concentrations; however, the file length will be increased dramatically. These can be easily figured in a spreadsheet. Multiply pounds per acre-feet by 0.3677 to get mg/L. The NPOL options may appear confusing, but are actually quite simple: NPOL is the TOTAL number of pollutants, including TS. But as TS needs no pollutant strengths, one less line of pollutant strengths must be read. Up to eight additional pollutant strengths may be read from either this land use file (+NPOL) or an external Pollutant File (—NPOL).

Ntitle	Npol	Output	EMC	Acres

Name	Value	Description
Ntitle	+	Number of title lines INCLUDING Project, RNfile/POLfile, and LandUse
Npol	0	Sediment washoff NOT simulated (runoffs only)
	1	Total event sediment washoff simulated
	+	NPOL—1 pollutant strengths (8 max) appear in LU File below.
	-1	Total Solids (TS) washoff will be summarized for EACH GROUP. Pollutant strengths can be modeled without more washoff simulations
	—	|NPOL|—1 pollutant strengths (8 max) appear in POL file. Options and default resets will also be read.

Output	0	Washoffs summarized for Standard summary of washoffs
	1	Extended summary of washoff, accumulation, sweeping, catchbasin sediment trapping and catchbasin cleaning events
	+	Only Accumulation on impervious surfaces summarized
EMC	±	Event Mean Concentrations are also listed in Washoff Summary
	0	EMC's are not listed (and file is MUCH shorter)
Acres	+	Acres of land use
	—,0	Defaults to 1.0, and results are normalized to per-acre

Title(1) / Project Title

The first title line is the Project Title which identifies all subsequent files. Later titles which differ from this will produce a warning message on the screen. Only the first 100 characters will be checked. Note that this file MUST have at least one title, and that the project title is included in the total number of titles above.

Title(2-N) / Other Titles

Any number of intermediate titles may follow, but only the first three will appear in output files. All lines are limited to 120 characters. Also, ALL titles, even those not echoed, must be included in the count.

Rnfile	Polfile	Rest of Title(N-1) / Other Title

The next-to-last title line contains information required when the land-use file is specified on the command line (thus the program is batch-run). The first element is the name of the rainfall event log from RAINEV that will drive the simulation. The second element will be the name of the pollutant file when the output selected requires pollutant strengths to be looked up here. Text following the last required element (first or second) will appear in output. Although up to 65 characters are allowed for filename and path, the line is still limited to 120 characters total. Note that if only one title line is present, it is used.

The last title line identifies the land use category with any control practices in place in subsequent output. Only the first 60 characters are retained, though up to 120 will be echoed in the output. Note that if fewer than three titles are present, the last title is used.

The first data line relates the duration of runoff to rainfall:

Arun	Brun	WetAccS	WetAccX

Name	Value	Description
ARun	0+	Hours constant in linear rainfall-runoff duration curve
BRun	+	Rate-of-change in linear rainfall-runoff duration curve
	—,0	BRun Defaults to 1.1, ARun to -0.6
WetAccS	+	Wet accumulation maximum rate (units per inch rainfall)
WetAccX	+	Wet accumulation minimum rainfall (inches)

EIA	ImpInt	IMax	IRate	PervInt	PMax	PRate

The second data line relates event rainfall depth to runoff volume for both impervious and pervious surfaces. The proportion of effective impervious area is followed by two sets of three variable loss parameters for impervious and pervious surfaces, respectively. Note that the rainfall depths are now in INCHES.

Name	Value	Description
EIA	0+	Effective Impervious Area
ImpInt	0+	Initial Loss (Lint) on impervious areas (inches)
ImpMax	0+	Maximum Variable loss (Lmax) for impervious areas
ImpRate	0+	Rate of approach to maximum loss (Lrate) for impervious areas
PervInt	+	Initial Loss on pervious surfaces
	0	Pervious contributions to runoff and washoffs are ignored
PervMax	0+	Maximum variable loss for pervious areas
PervRate	0+	Rate of approach to maximum loss on pervious areas

The third data line is required when total solids washoffs are simulated. It contains the parameters for accumulation, fraction of accumulation available for washoff, and for pervious washoff.

Curbdn	Sedint	Sedeq	Sedrate	SFMax	SFRun	Lperv	Rperv

Name	Value	Description
CurbDN	+	The number of curb feet per acre when Curb and Gutter (CG) washoff is simulated. Accumulation of sediment is in pounds per curb mile.
	0	Overland Flow (OF) washoff will be simulated; Accumulation of sediment is in pounds per impervious acre.
SedInt	0+	Initital sediment accumulated on impervious areas
SedEq	+	Equilibrium Sediment accumulation on impervious areas
SedRate	+	Rate of approach to SedEq (per day)
SFMax	+	Maximum fraction of accumulation available for washoff during an event
	0	Defaults to 1.0 (all accumulated sediment available)
SFRun	0+	The availability of accumulated sediment quadratically increasing, reaching SFMax with SFRun Inches of runoff (overall)
Lperv	0	Pervious washoff is not considered
	+	Maximum pounds-per-pervious acre that can wash off
Rperv	0+	Inches of pervious runoff, above which pervious washoff is constant

The fourth data line is also required when total solids washoff is to be simulated. These parameters transform the basin-wide hydrograph into the hydrograph at a stormwater inlet and allow hydraulic parameters such as depth of flow to be calculated. Note that the cross slope (Z) and the overland flow width share the same position for convenience.

Cbdn	Zwidth	Slope	Nvalue

Name	Value	Description
CBDN	+	Catchbasin Density: number of catchbasins per acre
Zwidth	+	Representative flow width for overland flow; gutter cross-slope for Curb and Gutter
Slope	+	Slope of energy gradeline (ie upland flow path)
Nvalue	+	Manning's N-value (0.08 Curb and Gutter, 0.12 overland flow)

The remaining lines containing information for other pollutants, control options, and default parameter resets, are identical for both the Land Use File and Pollutant File. These are described below under the Pollutant File. Note, though, that when the same control strategies or default resets are invoked more than once, only last appearance and/or the Land Use File value is used. Also, these lines can be commented out by placing comment characters (see above) anywhere on the line.

4.3 SIMPTM Pollutant and Option File (*.POL)

This file may be referenced by land-use files in SIMPTM, allowing often-used information to be maintained in only one place. Following the title block are up to eight pollutant strength lines having pollutant-to-solids mass fractions for as many as eight pollutants in addition to total solids. Optional control strategies and default resets which follow may appear in any order are superseded by appearances (if any) in the "calling" land use file. The subsequent option lines are ignored. Option lines may be commented out by placing ";","!", or "*" anywhere on the line. Again, input lines are limited to 120 characters.

The first line determines the arrangement of the remaining file, and selects the output options.

Ntitle	Npol

Name	Value	Description
Ntitle	+	Number of title lines INCLUDING Project
Npol	0	Only runoff duration and depth will be simulated (no pollutants or TS)
	1	Total Solids (TS) washoff will be summarized
	+	NPOL-1 (8 max) additional pollutants will be read (below) and summarized

The first title line is the Project Title which identifies all subsequent files. Later titles which differ from this will produce a warning message on the screen. Only the first 100 characters will be checked. Note that this file MUST have at least one title, and that the project title is included in the total number of titles above.

Any number of intermediate titles may follow but none appear in the output. ALL are counted in Ntitle, however.

4.4 Lines Common to both Land Use (*.LU) and Pollutant (*.POL) Files

Pollutant Strengths

The number of pollutant lines equals NPOL—1, as TS, the first pollutant, needs no pollutant strengths. More than the maximum of eight lines may appear; the NPOL from the "calling" file (Land Use or Basin) will be used. Ensure that the POL File Npol counts all pollutant strength lines, whether used or not, so that any lines which follow can be read properly. Also remember that pollutant names and order must be consistent throughout a project. On each line, the pollutant name (up to 16 characters are kept) is followed by eight pollutant strengths or mass fractions, one for each particle size group:

Pname	Pstr1	Pstr2	Pstr3	Pstr4	Pstr5	...	Pstr8

Name	Value	Description
Pname	A	Name of pollutant (must agree throughout project)
PSTR(1-8)	0+	EIGHT units of pollutant per pound of total solids for each of 8 groups

BMP Options

The remaining optional lines may appear in ANY ORDER, and are identified by a key word. These lines may be commented out as discussed above by either (;), (!), or (*) anywhere on the line. In general, there are two options:

- Invoking pollutant control strategies such as street cleaning or catchbasin traps (cleaned or not) and
- Resetting default parameters.

The control options usually require their own parameters.

Street Sweeping

Street sweeping is indicated by SWEEP followed by parameters describing the sweeping schedule and the average fraction of curb lane blocked by parked cars.

SWEEP	Swpfrq	Month	Day	Year	Hour	Park

Name	Value	Description
SwpFrq	0	Streets are not cleaned or swept regularly
	+	Number of days between scheduled sweepings (skipped during rainfall)
Month	1-12	Month of first scheduled sweeping
Day	1-31	Day of first scheduled sweeping
Year	+	Year of first scheduled sweeping
Hour	0-23	Hour of first scheduled sweeping
Park	0—1	Fraction of street blocked by parked cars (and not swept)

Sediment Trapping Catch Basins

To invoke sediment trapping catchbasins options TRAP must appear first on the line. Following it are the average dimensions of the sediment traps and the average depth of sediment at the start of the simulation.

Following these is the cleaning schedule and start day, if used. Again, if some parameters are not used, their places must still be filled with zeroes.

TRAP	Area	Hmax	Hint	Clnfrq	Month	Day	Year	Rhosed

Name	Value	Description
Area	+	Square feet of sediment trap base
HMax	+	Maximum Depth (ft) of sediment storage possible
Hint	+	Initial depth (ft) of sediment stored
ClnFrq	0	Sediment traps are not cleaned regularly
	+	Number of days between scheduled cleanings
Month	1-12	Month of first scheduled cleaning
Day	1-31	Day of first scheduled cleaning
Year	+	Year of first scheduled cleaning
RhoSed	+	Pounds DRY sediment per cubic foot WET sludge in the traps
	—,0	Defaults to 48

Internal Parameter Redefinitions

The default redefinitions involve sets of eight values for each of eight size groups. The current defaults appear in Table 4.1. Note that if even one of a set is re-defined, ALL must be.

ITEM	Value1	Value2	Value3	Value4	...	Value8

Parameter	Description of what is reset
PSDist	Resets distribution of sediment into eight groups. This is likely to be changed most often. Nevertheless, defaults were coded for convenience.
SSmin	Resets the base residual accumulations for each group, below which sweeping is ineffective. This parameter is only used when street sweeping is invoked.
SSeff	Resets the street sweeping removal efficiencies for each group. Again, this set is only used with street sweeping.
Atrap	Resets one of two coefficient sets used with the catchbasin trap efficiencies (see text). Reset only with caution, as very little data exists.
Btrap	Resets the other of two sets used with the catchbasin trap efficiencies. Again, reset only with extreme caution.
DIA	Resets the particle diameters for the eight groups simulated. This option has been added in case a need arises, but all documentation refers to these original sizes. Use of this option is strongly discouraged.

TABLE 5.1

Parameter Input File Layouts

RAINEV Instruction File (*.IN):

Line No.	Fields 1	2	3	4	5
1	Ntitle	Simptm	Output	Input	
2	Project				
3	RNTitle				
**					
4	Lapse	Trace	Thresh1	Thresh3	Thresh6

SIMPTM Land Use File (*.LU):

Line No.	Fields 1	2	3	4	5	6	7	8	9
1	Ntitle	Nprl	Output	EMC	Acres				
2	Project								
**	Titles								
3	RNFile	POLfile							
4	LandUse								
5	ARun	BRun	EvtfacS	AvtfacX					
6	Eia	Impint	ImpMax	ImpRate	Pervint	PervMax	PervRate		
opt. 7	CurbDN	Sedint	SedMax	SedRate	SFMax	SFRun	Lperv	Dperv	
opt. 8	CBDN	Zwidth	Slope	Nvalue					
**	Pname	Pstr1	Pstr2	Pstr3	Pstr4	Pstr5	Pstr6	Pstr7	Pstr8
		—— Optional, any order, last of each type used ——							
	"Sweep"	Swpfrq	Mo	Day	Yr	Hr	Park		
	"Trap"	Area	Hmax	Hint	Clnfrq	Mo	Day	Yr	RhoSed
	"Dia"	1.31E-04	3.18E-04	6.39E-04	1.27E-03	2.54E-03	5.09E-03	1.48E-02	2.29E-02
	"PSdist"	0.11	0.10	0.17	0.21	0.18	0.10	0.09	0.04
	"SSMin"	15	20	30	30	20	7	6	3
	"SSeff"	0.44	0.52	0.47	0.50	0.55	0.60	0.78	0.79
	"Atrap"	5.44	4.13	1.89	4.87	4.71	6.15	1.73	1.73
	"Btrap"	0.40	0.42	0.33	1.02	1.13	1.47	1.20	1.20

SIMPTM and SUMMRY Pollutant File (*.POL):

Line No.	Fields 1	2	3	4	5	6	7	8	9
1	Ntitle	Nprl							
2	Project								
**	Titles								
**	Pname	Pstr1	Pstr2	Pstr3	Pstr4	Pstr5	Pstr6	Pstr7	Pstr8

SUMMRY Basin Composition File (*.BAS):

Line No.	Fields 1	2
1	Project	
2	POLfile	
3	BasName	
4		
**	Acres	Iename(3)

NOTES:

1 An unlimited number of "**" lines may be inserted where shown

2 Fields are separated by commas or spaces, any length, but ALL must be filled

SECTION 5

APPENDICES

Contents at a glance

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APPENDIX A
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APPENDIX B
RUNTIME ERROR MESSAGES

The following table shows the error messages that may occur during execution of an F77L Version 5.00 compiled program and some explanation for messages which are not completely self-explanatory. The error number is the number which is displayed if the error message file (.EER) is not available to the runtime error processor. The following table lists the error codes, IOSTAT numbers, and the errors to which they refer. Note that messages are provided in English, French, German, Spanish, Italian, and Dutch. In order to change the language, replace F77l.EER with language.EER.

Error Number	Error Message Text
1	Insufficient RAM to Continue Execution
2	Program Stack Exhausted
3	NDP Divide by Zero
4	NDP Arithmetic Overflow
5	NDP Arithmetic Underflow
6	NDP Error - Invalid Number, Integer Overflow, or 0/0
7	Integer Divide Error
8	Integer*2 Overflow
9	Chain File Error File name given in CHAIN statement is invalid, does not exist, or is not an F77L-compiled ".EXE" file.
10	System Error During Chain Chain was unable to write either blank common or the FCB's for OPENed files to a temporary file, or memory blocks could not be adjusted.
No #	System Error After Chain The Chained-to program was unable to open and read the temporary file passed to it by Chain.
11	SUBROUTINE Subprogram Invoked as a FUNCTION
12	FUNCTION Subprogram invoked as a SUBROUTINE
13	Subprogram Argument Count Differs from Caller
14	Subprogram Alternate Return Count Differs from Caller
15	Substring Bounds are Poorly Defined

(continued)

Error Number	Error Message Text
16	Array Subscript Exceeds Allocated Area
17	Not Used
18	Not Used
19	DO Increment is Zero
20	Adjustable Array Dimension is Not Positive
21	Not Used
22	SQRT Argument Negative
23	Invalid Argument for MOD
24	Invalid Argument for INT
25	Invalid Argument Value for LOG
26	Invalid Argument Value for LOG10
27	Invalid Exponentiation
28	Invalid Exponentiation
29	Invalid Argument Value for SIN
30	Invalid Argument for COS
31	Invalid Argument Value for TAN
32	Invalid Argument Value for TANH
33	Invalid Argument Value for COSH
34	Not Used
35	Invalid Argument Value for ASIN Function
36	Invalid Argument Value for ACOS Function
37	Not Used
38	Invalid Argument Value for ATAN2 Function
39	Not Used
40	COMPLEX Divide by Zero
41	Not Used

(continued)

Error Number	Error Message Text
42	Runtime System Error in Subroutine "SYSTEM" An unexpected error occurred while invoking the subroutine SYSTEM.
43	Insufficient RAM for "COMMAND.COM" in Subroutine "SYSTEM"
44	"COMMAND.COM" Is Not Available in Subroutine "SYSTEM"
45-50	Not Used
51	Invalid Argument for COTAN
52	COTAN not defined for argument of multiple pi
53	Invalid argument for DCOTAN
54	DCOTAN not defined for argument of multiple pi
55	GAMMA not defined for x < -169.9 or x> 170.7
56	GAMMA not defined for zero or negative INTEGER
57	DGAMMA not defined for x < -169.9 or x> 170.7
58	DGAMMA not defined for zero or negative INTEGER
59	ALGAMMA not defined for x <= 0.0
60	DLGAMMA not defined for x <= 0.0
61	Invalid bit number specification in BTEST
62	Invalid bit number specification in IBSET
63	Invalid bit number specification in IBCLR
64	Inconsistent argument(s) in IBITS
65	Inconsistent argument(s) in MVBITS
66	Inconsistent argument(s) in ISHFT
67	Inconsistent argument(s) in ISHFTC
68-80	Not Used

The following errors may occur during execution of an input/output statement:

Error No.	IOSTAT No.	Error Message Text
81	1	Insufficient RAM to Continue Execution
82	2	Program Stack Exhausted
83	3	NDP Divide by Zero
84	4	NDP Arithmetic Overflow
85	5	NDP Arithmetic Underflow
86	6	NDP Error - Invalid Number, Integer Overflow, or 0/0
87	7	Integer Divide Error
88	8	Integer*2 Overflow
89		Not Used
90		Not Used
91	11	FORMAT Repeat Count Less Than 1 or Greater Than 32767
92	12	Invalid Direct File Record Number Record number must be positive
93	13	Data Transfer Beyond End of Record
94	14	Data Transfer Beyond End of File
95	15	System I/O Error An error in the I/O system which was caused by unexpected circumstances
96	16	Invalid Close Parameters Parameters on the CLOSE statement conflict with those on OPEN
97	17	Invalid or missing Record Length Specification Record length is not positive. Record length was not specified when required, or was specified for unformatted sequential
98	18	Invalid ACCESS = Specification Specification text not recognized
99	19	Invalid FORM = Specification Specification text not recognized or conflicts with other specifiers
100	20	Invalid STATUS = Specification Specification text not recognized
101	21	Invalid BLANK = Specification or used with unformatted file Specification text not recognized or is used with an unformatted file

(continued)

Error No.	IOSTAT No.	Error Message Text
102	22	Unable to Create File Disk or directory is probably full
103	23	Invalid Parameter Change on Open File Only CARRIAGE CONTROL =, RECL =, PAD =, DELIM = and BLANK = parameters may be changed when re-OPENing a file
104	24	Invalid I/O OPeration on a non-Disk Device DOS non-disk devices cannot be backspaced, rewound or closed
105	25	Invalid Syntax in FORMAT
106	26	FORMAT Specification Incompatible with Data Type
107	27	Too Many Conversion Digits Requested in FORMAT
108	28	Invalid CARRIAGE CONTROL = Specification Specification text not recognized or is used on a file that is not formatted sequential
109	29	Invalid FORMAT Field Width
110	30	Invalid Logical Input
111	31	Invalid Character Input
112	32	Invalid List-Directed Input
113	33	Invalid Numeric Input
114	34	Output Field Width Exceeded
115	35	Invalid Unit Number Unit numbers must be non-negative numbers less than 32768
116	36	Initiating I/O While Doing I/O Functions in an I/O list may not themselves perform I/O
117	37	Direct I/O on a Sequential File Attempting to OPEN a file or device that can only be accessed sequentially, or specifying a record number on a sequential file
118	38	Sequential I/O on a Direct File Attempting to OPEN a file that can only be accessed as a direct file with ACCESS = "SEQUENTIAL"
119	39	Unit Already Connected
120	40	Invalid Hollerith Constant in FORMAT
121	41	File is in an Inconsistent State Due to a previous error condition, I/O may not be performed on this unit

APPENDIX C
EIA COMPUTATION

Methodology for Estimating Effective Impervious Area for Natural, Partially Urbanized and Urban Watersheds Based on Published U.S.G.S. Data for Watersheds throughout the Metropolitan Areas of Portland and Salem, Oregon

By
Roger C. Sutherland, P.E.

Introduction

One of the most difficult and important parameters that must be estimated during the drainage master planning process is the **effective impervious area (EIA)** of a basin or subbasin of interest. **Effective impervious area (EIA) is the portion of the mapped impervious area (MIA) within a basin that is directly connected to the drainage collection system.** the EIA fraction includes street surfaces, paved driveways connecting to the street, sidewalks adjacent to curbed streets, rooftops which are hydraulically connected to the curb, and parking lots.

EIA is usually measured as a fraction of total basin or subbasin area. In traditional urban runoff modeling, the EIA for a given basin is usually less than that of MIA. However, in highly urbanized basins, EIA's can approach and equal MIA's.

The **EIA** of a basin is an important parameter in the rainfall to runoff process because it directly affects the volume of runoff. In hydrologic models like HEC-1, the EIA of a subbasin is the impervious area value that should be used on the LU, LM, LS, and LH precipitation loss records. Models like HEC-1 assume that no precipitation losses occur on impervious subbasin areas. Thus, the portion of the precipitation landing on the area specified as impervious will become direct runoff. In actuality, the precipitation falling on impervious areas which are not hydraulically connected to the drainage collection system will not result in direct runoff. The impervious area that does <u>not</u> contribute to runoff should be subtracted from the MAPPED impervious area to obtain the EFFECTIVE impervious area, which is used in the model.

Determination of Effective Impervious Area

The direct measurement of EIA is a tedious exercise which is rarely undertaken as drainage planning budgets can seldom afford the excessive cost. To actually measure the EIA of a basin, it would be necessary to catalog and evaluate the effectiveness of the hydraulic connection between each of the impervious surfaces and the major collector systems. This extremely time consuming exercise is impractical for most drainage planning and design activities.

If a basin is gaged, the effective impervious area can be estimated by employing a rainfall to runoff model like HEC-1 to calibrate the EIA parameter. This calibration is performed by fixing reasonable estimates of the precipitation loss components for the pervious portions of the basin and then adjusting the EIA fraction to fit computed to observed runoff volumes. The calibration process should be undertaken for several observed rainfall events, with the final estimate of EIA representing an average or weighted average of those values calibrated for each individual storm.

Unfortunately, in the real world, observed rainfall to runoff data are rarely available. Therefore, empirical equations must be developed to compute realistic values of EIA based on physical basin parameters which may be conveniently acquired. The **U.S.G.S.** developed estimates of EIA for over forty watersheds throughout the metropolitan areas of Portland and Salem, Oregon (Laenen, 1980 and 1983). Working with this data base, the **U.S.G.S.** also developed an empirical equation to estimate EIA as a function of MIA.

U.S.G.S. EIA Equation

The U.S.G.S. investigated the EIA parameter for the 42 drainage basins located throughout the Portland and Salem metropolitan areas. As part of their rainfall to runoff modeling, the U.S.G.S. optimized the EIA's for each of the drainage basins. The measured mapped impervious areas (MIA's) and the resulting optimized, modeled EIAs are presented in Table C.1.

It should be noted that the modeling technique used by the U.S.G.S. lumped all of the precipitation excess into a single optimized percentage of the basin area that was assumed to be contributing runoff. This optimized value was defined as the effective impervious area. A potential problem with this technique is that it will over-estimate the "true" EIA for values of MIA less than about 10 percent. However, for MIA values greater than 10 percent and less than 50 percent, it can provide reasonable estimates of the EIA parameter.

Working with these optimized fractions of EIA and their corresponding MIA's (ranging from 0 to 1), the U.S.G.S (Laenen, 1983) developed the following equation:

$$EIA = 0.036 + 0.43\ (MIA)$$

Equation (1) has been found to work well for MIA values greater than 10% and less than 50%. A problem with Equation (1) is that it provides unrealistic EIA values for MIA values less than about 10% and greater than 50%. In drainage master planning, one commonly deals with small subbasins (20 to 70 acres) in which the ultimately planned MIA may range from as much as 60% to 90%. The U.S.G.S. equation does not provide reasonable estimates in this range. Several other equations were developed, based upon the U.S.G.S. data, to serve this need.

Sutherland EIA Equations

An exponential equation was chosen to describe the relationship between MIA and EIA based upon the U.S.G.S. calibrated EIA's for all basins with MIA $\geq$ 0.04 as follows:

$$EIA = MIA^{x}$$

The EIA's and MIA's range between 0.00 and 1.00 (0 and 100%).

TABLE C.1

Watershed Characteristics for Basins Throughout the Metropolitan Portland and Salem, Oregon Area (Laenen, 1980 and 1983)

STATION NUMBER	STATION NAME	BASIN NO.	DRAINAGE AREA (MI^2)	MAP IMP AREA MIA	USGS MODEL EIA	TOTAL SEWERED AREA SA	TOTAL URBAN AREA UA	URBAN SEWERED AREA USA	STORAGE AREA ST	UA LOCAL DRAINAGE SYSTEM	MAJOR TRIBS	MAIN STEM	MIA + SA	BASIN CATEGORY
142580	KELLY CREEK	1	4.16	16	12	25.0	16	100	0.1	S	O	O	41	P1
144690	VANCOUVER SEWER OUTFALL	2	1.00	49	16	86.0	75	100	0.0	S	S	S	135	U2
206320	BEAVERTON CREEK	3	6.63	23	22	54.0	72	75	2.0	S	O	O	77	P2
206330	BEAVERTON CREEK TRIB.	4	0.21	19	15	75.0	58	100	1.1	S	S	O	94	U1
206470	BUTTERNUT CREEK	5	0.82	12	10	15.0	22	68	0.3	S	O	O	27	P1
206900	FANNO CREEK	6	2.37	32	17	57.0	87	66	0.0	S	O	O	89	P2
207800	SINGER CREEK	7	0.28	28	15	20.0	82	24	3.7	S	O	O	48	P1
210400	NOYER CREEK	8	2.04	6	6	1.0	7	14	0.0	O	O	O	7	N
211110	WILL R.(ROBINWOOD) TRIB.	9	1.03	10	4	0.0	31	0	0.0	O	O	O	10	N
211120	WILL R.(OAK GROVE) TRIB.	10	0.74	36	12	0.4	86	0	0.3	O	O	O	36	P1
211130	KELLOGG CREEK	11	2.42	22	15	5.0	65	8	7.0	O	O	O	27	P1
211301	TRYON CREEK TRIB.	12	0.36	32	25	29.0	88	33	0.6	S	O	O	61	P2
211450	JOHNSON CREEK TRIB.	13	0.21	16	7	14.0	41	34	0.0	S	O	O	30	P1
211500	JOHNSON CREEK	14	26.50	7	7	3.4	14	24	0.3	O	O	O	10	N
211604	N.W. 11TH-EVERETT SEWER	15	1.98	36	32	73.0	69	100	0.0	S	S	S	109	U1
211610	S.E. 9TH-MADISON SEWER	16	1.53	39	41	89.0	96	93	0.0	S	S	S	128	U2
211614	N.E. HANCOCK-FLINT SEWER	17	1.36	43	43	90.0	98	92	0.0	S	S	S	133	U2
211617	N.ALBINA-KIRKPATRICK SWR	18	0.95	44	22	88.0	94	94	0.0	S	S	S	132	U2
211618	N. VANCOUVER-OWRSN SWR	19	0.34	46	18	85.0	98	87	0.0	S	S	S	131	U2
211625	S.E. 27TH-BYBEE SEWER	20	0.77	26	28	91.0	93	98	0.0	S	S	S	117	U1
211630	S.E. 27TH-BELMONT SEWER	21	0.54	35	35	94.0	99	95	0.0	S	S	S	129	U2
211800	SALTZMAN CREEK	22	1.48	1	4	0.0	1	0	0.2	O	O	O	1	N
211950	VANCOUVER LAKE TRIB.	23	0.44	30	12	43.0	70	61	2.2	S	S	O	73	U2
213040	COUGAR CREEK	24	2.88	25	11	41.0	50	82	3.8	S	O	O	66	P2
190840	CROISAN CREEK	25	4.54	4	8	7.0	12	58	0.9	S	O	O	11	P1
190930	UPPER PRINGLE CREEK	26	2.93	2	4	0.0	0	0	0.9	O	O	O	2	N
190955	W.F. PRINGLE CREEK	27	3.16	30	18	44.0	73	60	2.5	S	O	O	74	P2
190960	CLARK CREEK	28	1.69	34	20	44.0	88	50	2.5	S	O	O	78	P2
190970	PRINGLE CREEK	29	12.60	22	15	21.0	45	47	4.5	S	O	O	43	P2
191440	BATTLE CREEK	30	5.56	2	3	0.0	0	0	3.9	O	O	O	2	N
191460	WALN CREEK	31	1.47	22	9	34.0	23	100	1.5	S	O	O	56	P2
192100	GLENN CR. @OCAKS FY.RD	32	2.51	8	6	8.0	12	67	0.3	S	O	O	16	P1
192120	GLENN CR. @ORCHARD HT.RD	33	3.31	10	8	11.0	20	55	0.3	S	O	O	21	P1
192150	GIBSON CREEK	34	0.54	2	3	0.0	0	0	0.1	O	O	O	2	N
192210	CLAGGETT CREEK	35	3.08	27	20	72.0	61	100	2.7	S	S	O	99	U1
192215	HAWTHORNE D. @D ST.	36	0.48	43	12	60.0	73	82	0.0	S	S	O	103	U2
192220	HAWTHORNE D. @SUNNYSIDE	37	0.80	53	28	76.0	82	93	0.1	S	S	O	129	U2
192225	HAWTHORNE D. @EASTGATE PK	38	1.40	45	25	86.0	84	100	0.4	S	S	O	131	U1
192230	HAWTHORNE D.@HYACINTH ST	39	1.68	43	23	88.0	84	100	0.4	S	S	O	131	U1
199655	L.PUDDING R.TR.@CORDON RD	40	0.79	15	12	36.0	51	71	9.9	S	O	O	51	P1
199855	L.PUDDING R.TR.@LARDON RD	41	0.27	1	2	0.0	0	0	1.5	O	O	O	2	N
200050	L.PUDDING R.TR.@KALE RD	42	0.75	20	8	66.0	60	100	7.4	S	O	O	86	P2

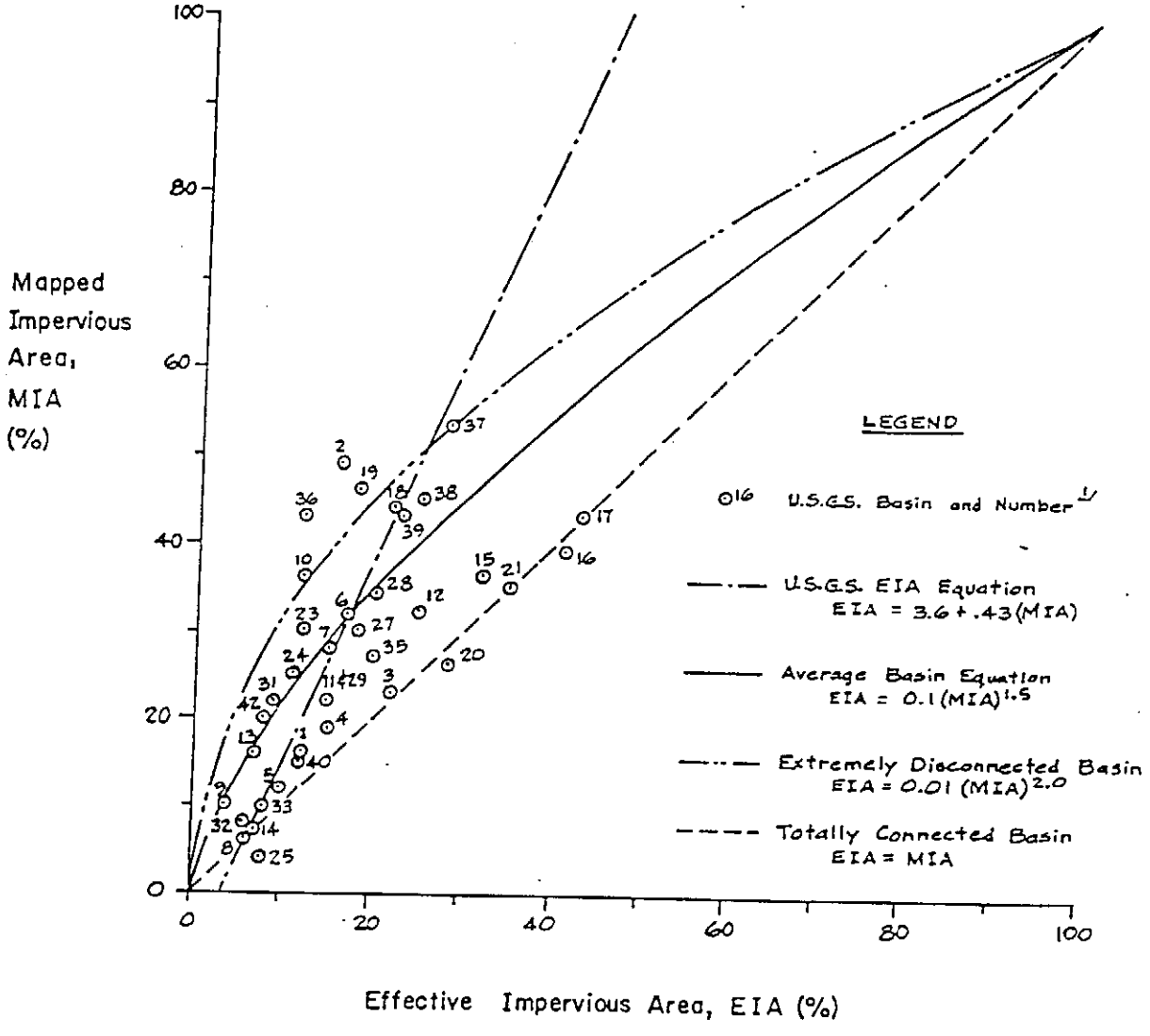
*NOTES O-OPEN S-SEWERED N-NATURAL P-PARTIALLY URBANIZED U-URBANIZED

While the entire data set was best predicted when x was 1.5, x varied depending upon subbasin conditions affecting EIA, primary the type of drainage system, as follows:

x	No.	Basin Description
1.0	I.	**Totally connected basins** where 100 percent of the urban area within the basin is storm sewered with all impervious surfaces appearing to be directly connected to the system.
1.2	II.	**Highly connected basins** where everything in Condition 1 applies except the residential rooftops are predominantly connected to the streets or storm sewer system.
1.5	III.	**Average basins** where the local collector systems for the urban areas within the basin are predominantly storm sewered with curb and gutter inlets, no dry wells or other drainage retention areas are known to exist, and the rooftops in the single family residential areas are not connected or piped directly to the street curb.
1.7	IV.	**Somewhat disconnected basins** where at least 50 percent of the urban areas within the basin are not storm sewered, but are served by roadside ditches, and the residential rooftops are not directly connected. Alternatively, Condition 1 may apply, but the basin is known to have a few dry wells or other retention areas.
2.0	V.	**Extremely disconnected** basins where only a small percentage of the urban rea within the basin is storm sewered, or a significant portion of the basin area drains to dry wells or other retention areas.

Figure C.1 compares the U.S.G.S. data and regression equation with the Sutherland EIA Equation for completely connected and disconnected and average conditions. The data variation demonstrates the difficulty of accurately estimating EIA. It is imperative that the drainage planner or engineer performs some degree of on-site investigation to determine which EIA equation may apply. The greatest strength of the **Sutherland EIA Equation** is its consistency in providing reasonable estimates of EIA for all values of MIA. Therefore, they can be used in the drainage planning process to estimate the change in EIA which will occur as a basin becomes urbanized.

FIGURE C.1



1/ EIA values were based on a U.S.G.S. rainfall to runoff model study. Only points with MIA ≥ 4 were plotted (Laenen, 1980 and 1983).